A NOVEL CYBER INFRASTRUCTURE ENABLING AI-DRIVEN ANALYTICS AND CONROL OF A POLYMER 3D PRINTER MACHINE

by

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A Thesis

Submitted to the
Department of Electrical and Computer Engineering
College of Engineering
In partial fulfillment of the requirement
For the degree of
Master of Science in Electrical and Computer Engineering
at
Rowan University
December 1, 2025

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Dedications

This work is dedicated to my family. Thank you for always supporting me in my endeavors. Each one of you has inspired me in this journey. You mean everything to me.

Acknowledgments

I would like to express my deepest gratitude to my thesis chair and advisor, Dr. Antonios Kontsos, for his invaluable guidance, support, and mentorship throughout this work.

I am also grateful to my thesis committee members Dr. Robi Polikar, Dr. Paromita Nath, Dr. Huaxia Wang, Dr. Christoforos Somarakis, Dr. Pete Mauro, and Dr. Richard Rabbitz for their time, thoughtful feedback, and contributions to my academic development.

My sincere appreciation also goes to the members of the Digital Engineering Hub (DEHub); Aymen Bahri, Essa Salem Al Amiri, Astrit Imeri, Evan Bretz, Ehsan Malekipour, Jai Hind Rathore, Alex Kinoian, Samuel Menaker, Nunna Sai Shreya and Varas Shah, whose collaboration and support enriched my research experience.

I would like to thank Vallen Systems for providing the necessary hardware and technical assistance, and Rowan University for providing the resources and research environment that made this work possible.

Finally, I am deeply thankful to my family and friends for their unwavering encouragement and support throughout this journey.

Abstract

Ahmed Ben Sidhom
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2025-2026
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Master of Science in Electrical and Computer Engineering

Additive manufacturing, particularly Fused Deposition Modeling (FDM), continues to face challenges with defect rates and process inconsistency. This thesis presents a real-time, multi-modal monitoring and autonomous control system designed to reduce human supervision and improve print quality and reliability.

The system combines three in-process monitoring modalities; acoustic emission sensing, computer vision–based defect detection, and thermal monitoring, synchronized with the printer's real-time position to localize anomalies within the printed geometry. A unified dashboard provides live visualization and logging for analysis.

A key contribution is the integration of a hierarchical LLM-powered multi-agent framework that interprets system data, queries historical logs, and directly controls the printer to enable closed-loop corrective actions.

Keywords: additive manufacturing, in-process monitoring, acoustic emission, computer vision, autonomous control, cyber-physical systems.

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List of Abbreviations

ABS	Acrylonitrile Butadiene Styrene
AE	Acoustic Emission
ACIC	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
CV	Computer Vision
CNN	Convolutional Neural Network
CPS	Cyber-Physical System
CSV	Comma-Separated Values
DAQ	Data Acquisition
DT	Digital Twin
ETL	Extract, Transform, Load
FIFO	First In, First Out
GPU	Graphics Processing Unit
HTTP	HyperText Transfer Protocol
IoT	Internet of Things
IoU	Intersection over Union
IP67	Ingress Protection rating 67
JSON	JavaScript Object Notation
LLM	Large Language Model
MAS	Multi-Agent System

List of Abbreviations (Continued)

mAP	mean Average Precision
MLP	Multi-Layer Perceptron
MTConnect	Machine Tool Connectivity standard
NPU	Neural Processing Unit
ONNX	Open Neural Network Exchange
PETG	Polyethylene Terephthalate Glycol
PLA	Polylactic Acid
RMS	Root Mean Square
RNN	Recurrent Neural Network
SFTP	Secure File Transfer Protocol
SQL	Structured Query Language
SSH	Secure Shell
STL	Stereolithography (file format)
TCP	Transmission Control Protocol
UI	User Interface
XML	eXtensible Markup Language
YAML	YAML Ain't Markup Language

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Chapter 1

Motivation and Research Overview

1.1 Introduction to Additive Manufacturing

Additive manufacturing (AM), commonly known as 3D printing, represents a transformative paradigm in modern manufacturing, enabling the layer-by-layer construction of complex geometries from digital models. Unlike traditional subtractive methods, such as milling or turning, which remove material from a solid block, AM builds objects by depositing material precisely where needed, minimizing waste and allowing for unprecedented design freedom. This technology has gained significant traction across industries, including aerospace, automotive, healthcare, and consumer goods, due to its ability to produce customized parts on-demand, reduce lead times, and facilitate rapid prototyping[1].

Among the various AM techniques, Fused Deposition Modeling (FDM) stands out as one of the most accessible and widely adopted processes, particularly for polymer-based materials. In FDM, a thermoplastic filament is fed through a heated nozzle, melted, and extruded onto a build platform in successive layers that solidify upon cooling as shown in Figure 1. The process relies on precise control of parameters such as nozzle temperature, extrusion rate, layer height, and print speed to achieve desired outcomes. FDM's popularity stems from its low cost, ease of use, and compatibility with a range of materials like PLA, ABS, and PETG. According to market projections, the global additive manufacturing sector is expected to grow by nearly 24% between 2023 and 2025[2, 3], with the 3D printing market reaching substantial economic value, underscoring its pivotal role in future production landscapes.

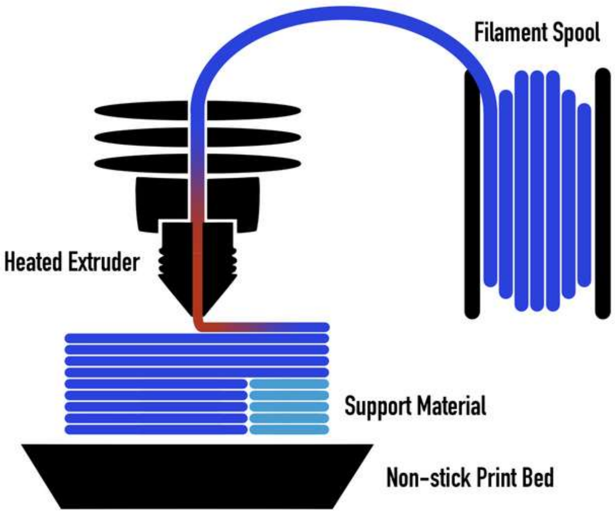

Figure 1. Schematic diagram of FDM process[4]

1.2 Challenges in FDM 3D Printing

Despite its advantages, FDM 3D printing is plagued by persistent challenges that hinder its reliability and widespread industrial adoption. High defect rates and poor process repeatability often result in failed prints, material waste, and the need for constant human supervision. Studies indicate that failure rates in desktop FDM systems can exceed 20%, with some observations reporting up to 41.1% of prints encountering issues[5]. These failures not only increase costs but also limit scalability, especially in production environments where consistency is paramount.

Common defects in FDM (Figure 2) include warping, where the printed part curls or deforms due to uneven cooling and thermal stresses; stringing, characterized by thin strands of material left between non-contiguous print areas; under-extrusion, resulting in gaps or weak layers from insufficient material deposition; over-extrusion, leading to excess material buildup and surface irregularities; layer shifting, caused by mechanical misalignment or vibrations; and spaghettification, where unsupported structures collapse into tangled filaments. Other issues encompass cracking (delamination) between layers, poor bed adhesion, and dimensional inaccuracies[6]. These defects arise from a complex interplay of factors, including material properties, environmental conditions (e.g., humidity and temperature), printer calibration, and slicing software settings. For instance, warping is exacerbated by differential cooling rates, while stringing often stems from inadequate retraction settings or high nozzle temperatures[7].

In addition to defects, FDM processes generate significant waste, with estimates showing 10-15% material loss from supports, failed prints, and post-processing[8]. Voids, surface roughness, and weak interlayer bonding further compromise mechanical integrity, making FDM parts anisotropic and less suitable for load-bearing applications without enhancements. These challenges underscore the limitations of open-loop systems, where printers operate without real-time feedback, perpetuating errors until post-print inspection.

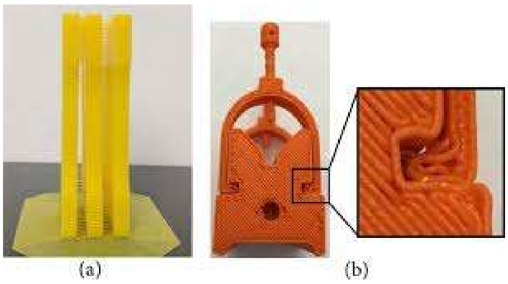

Figure 2. Common defects: (a) Stringing; (b) Under extrusion[9]

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1.3 The Need for Advanced Monitoring and Control

To mitigate these issues, there is a growing imperative for closed-loop monitoring systems in 3D printing. Such systems incorporate real-time sensing and feedback mechanisms to automatically detect anomalies during the build process and adjust parameters dynamically, thereby enhancing part quality and process efficiency. Benefits include reduced material waste, improved reliability, and the ability to correct defects in situ, preventing complete failures. For example, integrating sensors for thermal monitoring can provide feedback to maintain optimal temperatures, while computer vision enables immediate identification of visible defects like stringing or warping.

Closed-loop approaches have demonstrated significant improvements, such as achieving layer-wise corrections that enhance tolerance and reliability[10, 11]. In advanced setups, AI-augmented systems enable predictive monitoring and autonomous interventions, particularly beneficial for large-scale or complex prints. However, many consumer-grade printers still operate in open-loop modes due to the lack of integrated sensors, highlighting a gap in accessible, retrofit-friendly solutions[12].

1.4 The role of AI in Additive Manufacturing

Artificial Intelligence (AI) revolutionizes quality control in AM by enabling predictive analytics, defect detection, and process optimization. AI applications include generative design for complex geometries, real-time monitoring via computer vision, and machine learning models for anomaly prediction. In FDM, AI can analyze sensor data to adjust parameters like extrusion rates or temperatures, reducing defects and ensuring compliance with standards[13].

Machine learning models, such as convolutional neural networks (CNNs) for image-based defect classification or recurrent networks for time-series analysis, have achieved high accuracy in identifying issues such stringing (up to 100% in tests) and under-extrusion[14]. Furthermore, AI facilitates post-process analysis for continuous improvement, transforming raw data into actionable insights.

1.5 Industry 4.0, Industry 5.0, and Cyber-Physical Systems in 3D Printing

The integration of additive manufacturing (AM) into Industry 4.0 frameworks leverages cyber-physical systems (CPS) to create smart, interconnected manufacturing environments. In this context, CPS in 3D printing involves fusing physical printers with digital twins, real-time data analytics, and networked controls to enable autonomous operations, adaptive process control, and remote monitoring[15]. This integration supports predictive maintenance, optimized workflows, enhanced traceability, and rapid decision-making by aligning with Industry 4.0 pillars such as IoT connectivity, big data, and AI-driven optimization[16].

In AM, adopting both Industry 4.0 and 5.0 approaches as illustrated in Figure 3 means extending CPS to not only collect and analyze real-time sensor data but also integrate human expertise and decision support directly into the production loop. Examples include cloud-based CPS platforms for process parameter prediction, intelligent decision support systems that automate portions of the print workflow while allowing operator intervention, and human-machine co-design tools that facilitate mass customization. These synergies reduce human intervention where appropriate, boost operational efficiency, and enable scalable AM deployment in smart factories while ensuring that human intelligence and sustainability remain central to system design[17].

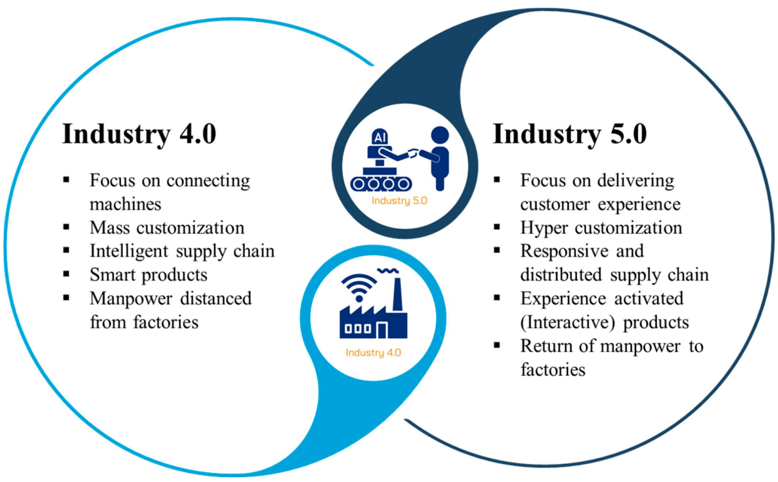


Figure 3. Highlights of Industry 4.0 and Industry 5.0[18]

1.6 Research Motivation

The motivation for this research stems from the identified deficiencies in current FDM systems, such as frequent failures from defects like under-extrusion, stringing, and spaghettification, which impede reliability and efficiency. These can be addressed through closed-loop monitoring that detects anomalies and adjusts parameters in real-time. Key gaps include the lack of standardized workflows for multi-modal sensing integration with controls, absence of effective data structures for AI tools, limited hierarchical multi-agent AI for natural-language interaction and autonomous intervention, and insufficient understanding of digital twins for continuous learning based on local parameters.

1.7 Research Overview

This thesis presents a novel cyber infrastructure for AI-driven analytics and control of a polymer 3D printing machine. The system integrates acoustic emission (AE) analysis with a Keras-based classifier and computer vision via YOLOv5 for defect detection. Also, thermal monitoring through OctoPrint. Data streams are synchronized with print head positions for spatial defect mapping. A Streamlit dashboard provides real-time visualizations, while an SQLite database enables persistent storage. A hierarchical LLM-powered multi-agent system, using Smolagents and Qwen-3-Coder, facilitates natural-language queries, database interactions, and autonomous controls, advancing toward digital twins and cyber-physical operations.

1.8 Thesis Organization

The remainder of this thesis is structured as follows: Chapter 2 reviews related background on AM, defects, monitoring techniques, and AI in manufacturing. Chapter 3 outlines research objectives and technical approaches. Chapter 4 details the implementation of monitoring subsystems. Chapter 5 covers database and agent-based control. Chapter 6 presents results and discussion. Chapter 7 concludes with future work.

Chapter 2

Related Background

2.1 Additive Manufacturing and Fused Deposition Modeling

The global AM industry has experienced steady growth, with the Wohlers Report 2025 indicating a 9.1% increase from $21.8 billion in 2024, driven by advancements in materials, software, and hardware. 2025 projections suggest continued expansion, with the market potentially reaching $26 billion, fueled by increased adoption in mass production and supply chain resilience. FDM accounts for approximately 68% of the global desktop and prosumer installed base in 2025, dominating the industrial 3D printer category with a 68.6% share within its subsegment[19]. This prevalence is attributed to its affordability and versatility, though challenges in scalability for high-volume production persist.

A key enabler of these advancements is the role of data structures in AM, which facilitate efficient representation, processing, and management of digital models throughout the manufacturing workflow. Common data formats like STL (StereoLithography) use triangular mesh structures to approximate 3D geometries, enabling precise slicing into layers for printing, though they lack support for color or multi-material information. More advanced formats, such as 3MF and AMF, incorporate hierarchical data architectures that include metadata on materials, textures, and build parameters, improving interoperability and reducing errors in complex designs[20]. Octree-based structures, for instance, allow adaptive slicing by subdividing space into variable-resolution voxels, optimizing print paths, minimizing material use, and enhancing resolution in detailed areas while speeding up computation for simpler regions[5]. Standardized data architectures, as proposed in recent frameworks, organize information across project management, feedstock materials, processing steps, measurements, and simulations, ensuring traceability and data integration for quality control and repeatability in AM processes.

Figure 4 shows, despite its advantages, FDM is prone to inconsistencies due to parameter sensitivity. Studies from 2014–2015, such as those by Turner et al. and Mohamed et al.[21, 22], highlight how variations in these parameters can lead to defects, affecting mechanical properties such tensile strength and dimensional accuracy. More recent data from 2023–2025 indicate failure rates in unsupervised FDM printing ranging from 15–50%, with economic losses estimated at $4.8 billion annually across global maker and professional communities[23]. These failures often result from material inconsistencies, mechanical issues, or suboptimal settings, underscoring the need for advanced monitoring to enhance reliability.

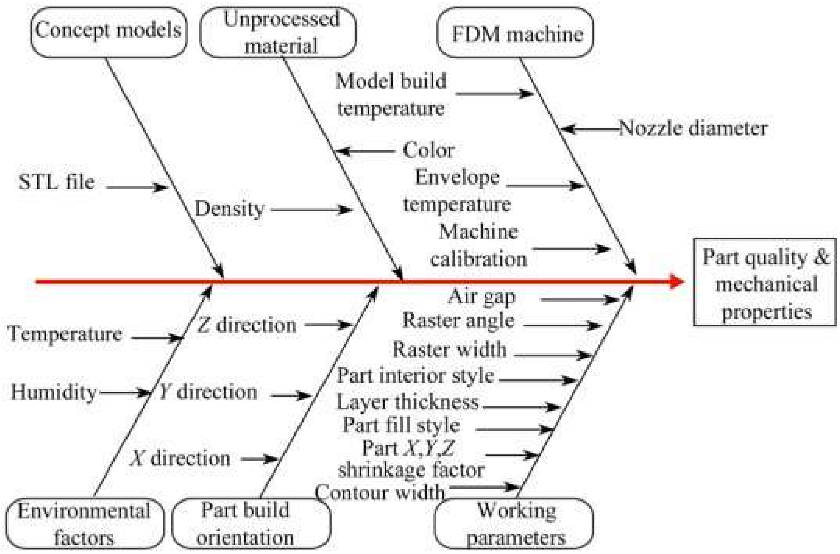


Figure 4. Cause and effect diagram of FDM process parameters[21]

2.2 Common Defects in FDM Printing

FDM printing is susceptible to a variety of defects that compromise part quality, structural integrity, and functionality (see Table 1). These issues arise from improper parameter settings, material properties, environmental conditions, or hardware malfunctions[5, 24, 25]. Common defects include:

- Warping and Curling: Occurs due to uneven cooling and thermal contraction, causing the edges of the print to lift from the bed. This is exacerbated by high printing temperatures or inadequate bed adhesion.
- Stringing or Oozing: Thin strands of filament left between non-contiguous parts, resulting from insufficient retraction or high nozzle temperatures.
- Under-Extrusion: Gaps or weak layers from insufficient material deposition, often caused by clogged nozzles, low extrusion multipliers, or filament diameter inconsistencies.
- Over-Extrusion: Excess material buildup leading to blobs, zits, or rough surfaces, typically from high flow rates or incorrect filament settings.
- Layer Shifting: Misalignment of layers due to mechanical issues like loose belts, vibrations, or stepper motor skips.
- Spaghettification: Tangled filaments from unsupported structures or failed layers, common in complex geometries without proper supports.
- Cracking (Delamination): Separation between layers from poor interlayer bonding, influenced by low temperatures or rapid cooling.
- Pillowing and Gaps in Top Layers: Holes or uneven surfaces on top due to insufficient infill or cooling fan issues.
- Overheating and Blobs: Localized melting or deformities from excessive heat retention, especially in small features.
- Z-Wobble: Wavy surfaces from Z-axis instability, often linked to lead screw misalignment.

These defects not only waste material and time but also necessitate reprints, contributing to high failure rates. Comprehensive troubleshooting guides emphasize calibration, material selection, and environmental control as key mitigation strategies.

Table 1

Common defects posed by the FDM printing[26]

Defect	Size range	Spatial topology	Category	Location
Shrinkage	Centimeteric	3D	Deformational	Combined
Warping		3D		Combined
Layer shifting		3D		Combined
Delamination		2D		Surface/internal/combined
Curling		3D		Combined
Gaps	Centimeteric/ millimetric	3D/4D	Deviations from the normal amount of material	Surface/internal/combined
Stringing		1D		Surface
Over-extrusion	Millimetric	2D/3D		Surface/internal
Under-extrusion		2D/3D		Surface/internal
Banding		3D		Surface
Cracks	Millimetric / micrometric	1D/2D/4D	Deformational	Surface/internal
Blobs		0D	Deviations from the normal amount of material	Surface
Voids	Micrometric/ nanometric	0D/4D		Surface/internal
Molecular cleavage	Angstrom	0D	-	Surface/internal/combined

2.3 In-Process Monitoring Techniques for FDM

In-process monitoring is essential for detecting defects in real-time, enabling corrective actions and improving overall print quality. Various techniques have been developed, leveraging sensors and data analysis to address FDM's challenges.

2.3.1 Computer Vision-Based Monitoring

Computer vision (CV) is the most explored modality for FDM monitoring, using cameras to capture images or videos of the printing process for defect detection as depicted in Figure 5. Early methods relied on simple thresholding and blob detection, but deep learning advancements, such as convolutional neural networks (CNNs) and YOLO models, have enhanced accuracy[27]. For instance, YOLOv5 and YOLOv4 Tiny have been employed for real-time detection of defects like stringing, spaghettification, and extrusion issues, achieving up to 90% mean average precision (mAP) with low latency. Systems integrate webcams or high-resolution cameras to analyze layer alignment, surface irregularities, and material buildup, often triggering pauses or alerts upon anomaly detection[28].

Limitations include dependency on lighting conditions, camera positioning, and enclosure transparency, which can obscure internal defects like voids or early delamination. Recent studies propose multi-head neural networks for generalizable error detection, trained on parameter deviations to improve robustness across printers.

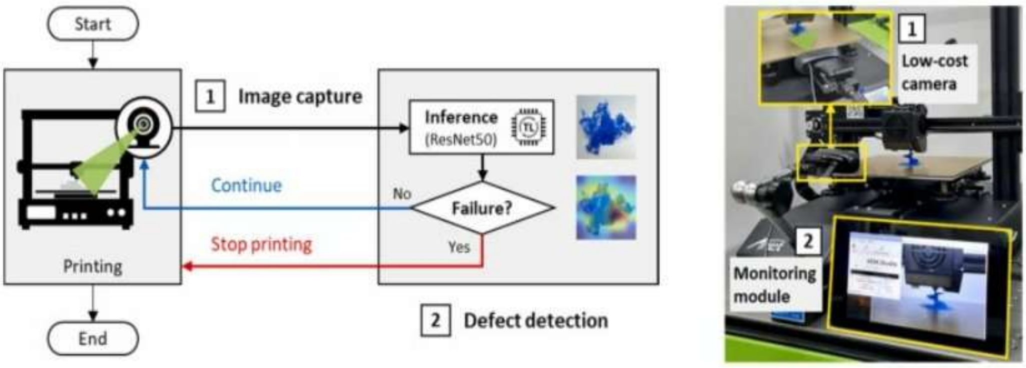


Figure 5. Computer Vision-Based Monitoring[29]

2.3.2 Acoustic Emission Monitoring

Acoustic emission (AE) monitoring detects high-frequency stress waves (1–10 MHz) generated by material deformations or defects during printing, offering sensitivity to internal anomalies without line-of-sight requirements. Sensors mounted on the nozzle or frame (Figure 6) capture signals associated with cracks, delamination, or extrusion irregularities. Machine learning models analyze features like amplitude, energy, and frequency to classify defects with accuracies up to 85%[27].

Advantages include real-time detection of hidden issues, such as micro-cracks or over-extrusion, but challenges involve high sampling rates, noise from stepper motors, and sensor coupling. Studies integrate AE with other modalities for comprehensive monitoring, such as warpage detection in FDM.

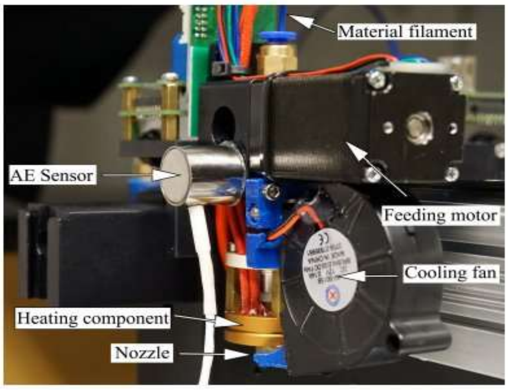


Figure 6. Acoustic Emission Monitoring[30]

2.3.3 Thermal Monitoring

Thermal monitoring uses infrared cameras or thermocouples to track nozzle, bed, and layer temperatures as seen in Figure 7, ensuring optimal adhesion and preventing defects such warping. Real-time systems provide feedback for closed-loop control, with resolutions of 0.1°C and accuracies of ±1–2°C. Studies show temperature variations affect extrusion quality, with slower speeds increasing heat buildup[28]. Integration with APIs like OctoPrint allows polling every second for trend analysis.

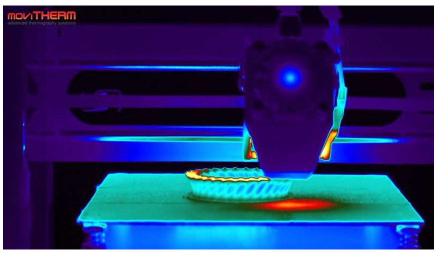


Figure 7. Thermal Monitoring[31]

2.3.4 Multi-Modal Fusion

Multi-modal fusion, illustrated in Figure 8, combines data from multiple sensors (e.g., CV, AE, thermal) to enhance detection accuracy and robustness. Approaches such decision-level fusion and unsupervised methods inspired by CLIP models integrate heterogeneous data for anomaly classification[32, 33]. This enables inter-layer flaw detection in powder bed fusion and FDM, improving process monitoring by leveraging complementary strengths (e.g., CV for visible defects and AE for internal ones). Challenges include data synchronization and computational complexity, but benefits encompass reduced false positives and better decision-making[34].

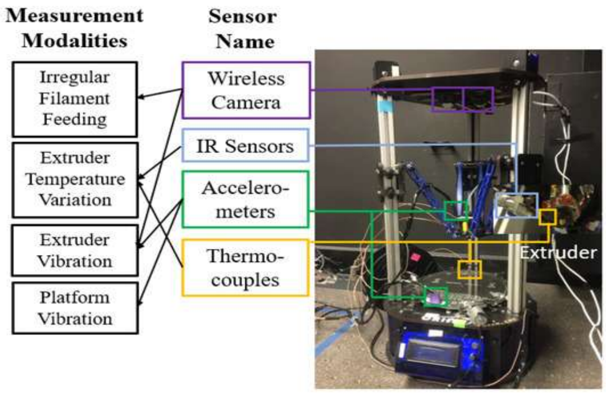


Figure 8. MOSS—Multi-Modal Best Subset Modeling in Smart Manufacturing[35]

2.3.5 Spatial Registration of Defects

Spatial registration maps detected defects to precise 3D coordinates within the printed geometry (see Figure 9), using G-code parsing and sensor synchronization. Techniques employ optical coherence tomography (OCT) or mesh metrology for fidelity assessment, achieving errors under 6.5 mm[36]. This enables root-cause analysis and proactive interventions, with machine learning aiding localization without registration.

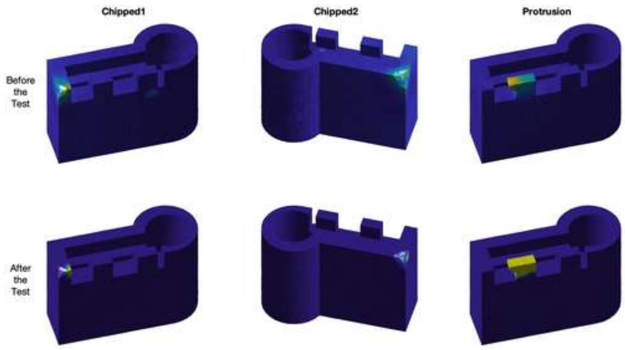


Figure 9. Defective Parts with Significant Points Highlighted in Yellow[37]

2.3.6 OctoPrint Ecosystem and Extensions

As shown in Figure 10, OctoPrint is an open-source web interface for remote printer control and monitoring, supporting plugins for enhanced functionality. Key extensions include PrintWatch for AI-driven defect detection, Obico for failure alerts, and SimplyPrint for farm management. The ecosystem facilitates temperature polling, G-code upload, and live feeds, with over 300 plugins available for customization[38].

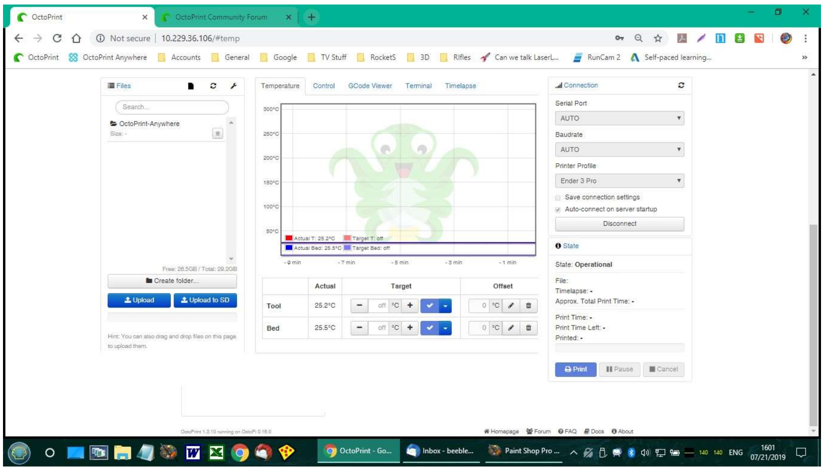


Figure 10. Customizing Octoprint Display

2.4 Large Language Models and Multi-Agent Systems in Manufacturing

Large Language Models (LLMs) and multi-agent systems (MAS) are emerging as pivotal tools in manufacturing, enabling intelligent automation and decision-making. LLMs such Qwen-3-Coder facilitate natural-language interactions for process control, while MAS involve collaborative agents for tasks such as scheduling and quality assurance (see Figure 11). In 2025, LLM-based MAS architectures, such as those using Claude or GPT variants[39, 40], support shopfloor intelligence, with frameworks such AutoGen and LangChain[41, 42] enabling multi-agent collaboration. Applications include adaptive manufacturing systems where agents query databases, control equipment, and perform root-cause analysis, reducing human intervention. Challenges encompass integration with legacy systems and ensuring security, but benefits in efficiency and scalability are significant.

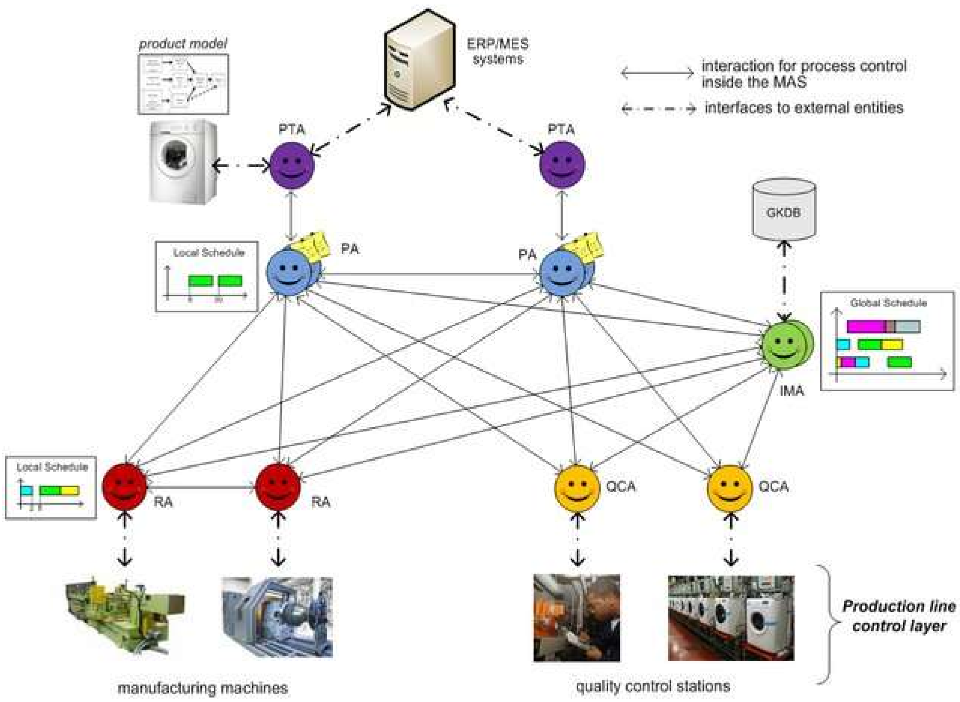


Figure 11. Multi-agent System Architecture for the Production Line.[43]

2.5 Commercial and Closed-Source Solutions

Commercial solutions for FDM monitoring include Deep Space Monitoring by 3D Systems (Figure 12) for real-time metal printing analysis[44], Addiguru for layer-by-layer defect detection[45], and Phase 3D for retrofit quality assurance. EOS's in-process systems focus on powder bed fusion, while 3DQue's AutoFarm handles print farms with auto-removal[46]. Closed-source tools like INTAMSYS software offer material versatility, but limitations include high costs and lack of customization compared to open-source alternatives[47].

Figure 12. Deep Space | 3D Systems[48]

Chapter 3

Research Objective and Technical Approach

3.1 Research Gaps Identified

The field of AM, particularly FDM, has seen significant advancements in monitoring and control technologies. However, several critical gaps persist that limit the transition from open-loop to fully autonomous, closed-loop systems. These gaps, identified through chapter 2 comprehensive review of existing literature and industry practices as of 2025, are outlined below and form the foundation for the objectives of this research.

First, there is a lack of standardized workflows to allow multi-modal sensing to be integrated with equipment controls to achieve near real-time defect detection and control. While individual sensing modalities such computer vision (CV) and acoustic emission (AE) have been explored, their fusion into cohesive systems remains fragmented. For instance, a 2025 study on enabling multimodal sensor fusion in AM highlights the absence of uniform methodologies that comprehensively address data synchronization, processing latency, and control feedback loops[49, 50]. This results in systems that are either overly specialized or inefficient, leading to delays in defect mitigation and increased failure rates. Recent analyses indicate that without standardized protocols, multi-modal systems often suffer from interoperability issues, with integration challenges exacerbating up to 30% of operational inefficiencies in FDM setups.

Second, there is an absence of a data structure to enable effective ways to develop AI-based tools for detection, identification, and control of 3D printing systems. Current databases in AM monitoring are typically siloed, lacking the multimodal persistence needed for retrospective analysis and model retraining. A key issue is the handling of heterogeneous data streams (e.g., time-series AE signals[51], image-based CV outputs, and thermal readings), which often leads to data loss or misalignment. Research from 2025 on digital twins for defect detection in FDM emphasizes that without robust, schema-aware databases, AI tools struggle with scalability, resulting in models that underperform in variable environments[52, 53]. This gap is compounded by the need for data structures that support continuous learning, where local process parameters (e.g., filament type or ambient conditions) can dynamically influence AI decisions.

Third, there are limited ways to integrate hierarchical multi-agent AI for natural-language interaction and conditional autonomous intervention in 3D printing systems to enable cyber-physical system (CPS) operations. While large language models (LLMs) have been applied to manufacturing, their use in hierarchical agents for AM is nascent. A 2025 framework on LLM-3D print demonstrates potential for emergent reasoning in defect resolution but notes the scarcity of architectures that combine natural-language querying with autonomous control[13]. Existing systems often rely on rigid rule-based interventions, lacking the conditional logic needed for complex scenarios, such as adapting to evolving defects such progressive delamination. This limits CPS capabilities, where physical printers must seamlessly interface with digital agents for real-time autonomy.

Finally, there is a lack of understanding on how digital twins for manufacturing systems could be developed to allow continuous learning and adaptation based on local process parameters and variable inputs and outputs. Digital twins (DTs) promise virtual replicas for simulation and prediction, but in FDM, their implementation is hindered by incomplete sensor integration and data fidelity[50]. A critical review from 2025 on DT-enabled model predictive control in AM identifies gaps in handling variable inputs (e.g., material variability) and outputs (e.g., part quality metrics), with current DTs often failing to achieve sub-100 ms synchronization for real-time adaptation[53]. This underscores the need for foundational research to bridge simulation with physical processes, enabling proactive defect prevention.

These gaps collectively contribute to persistent high failure rates (15–50%) in unsupervised FDM printing, as noted in broader AM trend reports for 2025[19]. Addressing them requires an integrated approach that combines hardware, software, and AI innovations.

3.2 Research Objectives

To address the identified gaps, this thesis establishes a set of targeted research objectives aimed at advancing AI-driven analytics and control in polymer 3D printing. These objectives build on emerging trends in AI for AM, such as generative design and process optimization, as highlighted in 2025 executive surveys. Each objective is designed to contribute to a cohesive cyber infrastructure, with implications for Industry 4.0 scalability.

The first objective is to develop synchronized, real-time data acquisition pipelines for the use of multiple added sensors to commercial 3D printing systems. This involves creating data infrastructure that handle data from CV, AE, and thermal sensors with minimal latency (target <50 ms), ensuring alignment with printer coordinates. Such synchronization is crucial for multivariate sensing, enabling comprehensive defect monitoring beyond single-sensor limitations. By retrofitting commercial printers such the Ender 3 , this objective democratizes advanced monitoring, aligning with 2025 goals for AI-enhanced efficiency in AM.

The second objective is to implement high-accuracy deep learning models for defect classification in 3D printing. Focusing on models such YOLOv5 for CV-based detection and Keras neural networks for AE signal classification, this aims for accuracies exceeding 85–90% in identifying defects such as under-extrusion and stringing. Training on induced defects ensures model robustness, with real-time inference supporting immediate interventions. This objective advances AI applications in quality control, reducing waste by up to 20%.

The third objective is to achieve low-latency spatial registration of defects by real-time parsing of execution and sensing databases. By parsing G-code logs and synchronizing with sensor timestamps, defects are mapped to XYZ coordinates with errors ≤6.5 mm. This enables precise localization, facilitating targeted corrections and post-print analysis. In 2025 contexts, this supports predictive maintenance in AM, where spatial data informs DT simulations.

The fourth objective is to create a persistent multimodal database enabling retrospective analysis and continuous model improvement. Using SQLite with normalized schemas (e.g., tables for print jobs, CV metadata, and AE time-series), this database stores features like waveforms and bounding boxes for querying and retraining. It addresses data structure gaps, enabling ML pipelines for iterative enhancements.

The fifth objective is to design and integrate a hierarchical LLM-powered multi-agent architecture capable of natural-language interaction and conditional autonomous control. Leveraging models such Qwen3-Coder, agents handle SQL queries, printer controls, and reasoning, enabling user commands like "pause if defects exceed threshold." This fosters CPS operations, with 2025 advancements in agentic coding enhancing autonomy.

The sixth objective is to allow future development of digital twins for additive manufacturing. By laying groundwork through data pipelines and agents, this enables DTs for simulation-based learning, adapting to variables like material specs.

These objectives collectively aim to reduce human supervision and improve part reliability.

3.3 Technical Approach

The technical approach integrates hardware, software, and data flow to realize the objectives, drawing from established AM practices and 2025 innovations.

3.3.1 Hardware Components

The system is built around a Creality Ender 3 V2 FDM printer, modified with sensors for multi-modal monitoring as shown in Figure 13. Key components include:

- 3D Printer: Ender 3 V2 (build volume 220×220×250 mm, nozzle 0.4 mm, max speed 180 mm/s), chosen for its affordability and modifiability.

- Acoustic Emission Sensor: VS600-Z1 with Linwave/SpotWave DAQ for high-frequency (1–10 MHz) signal capture, mounted on the nozzle for internal defect detection.
- Webcam: Logitech C270 for CV, providing frames at 1 Hz.
- Host Computer: AMD Ryzen AI Max+ 395 processor with unified memory architecture (192 GB LPDDR5X shared between CPU, GPU, and NPU). The unified memory eliminates PCIe bottlenecks during simultaneous YOLOv5 inference, AE feature extraction, and Qwen3-Coder-30B agent execution.
- Raspberry Pi 4: Hosts OctoPrint, streams data via SSH/WireGuard.
- Temperature Sensors: Integrated via OctoPrint API for bed/nozzle readings.

These components enable full retrofit integration on a consumer-grade printer while leveraging the unified-memory AMD Ryzen AI Max platform as the central processing node for real-time multi-modal fusion, deep-learning inference, and LLM-powered agent orchestration.

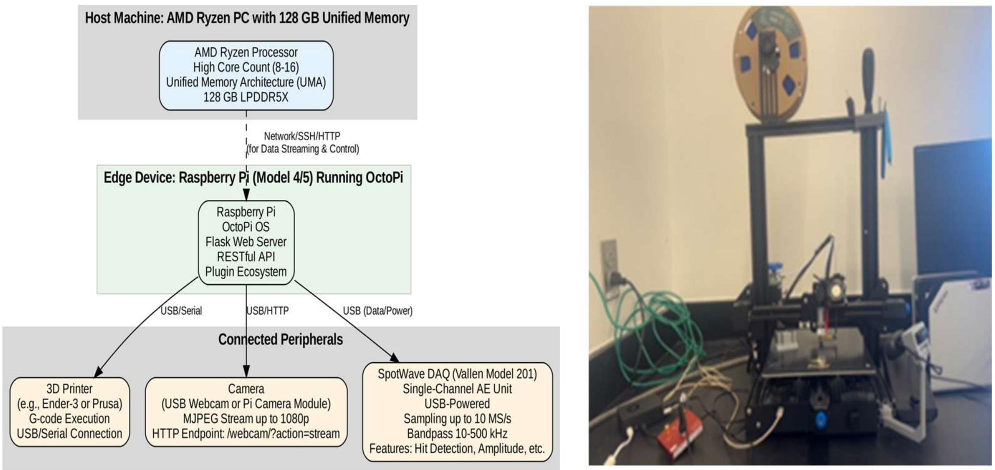


Figure 13. Hardware Components

3.3.2 Software Components

The software components are illustrated in Figure 14, they are modular; Focusing on monitoring, data management, UI, and AI.

- Monitoring Modules:

  AEMonitor: Streams AE data, extracts features (amplitude, energy), classifies via Keras (85% accuracy), maps to coordinates, plots signals, auto-pauses after 5 defects.

  CVMonitor: Captures frames, runs YOLOv5 (90.2% mAP@0.50, ~100 ms latency), annotates defects, auto-pauses after 3.

  TemperatureMonitor: Polls OctoPrint every 1s for trends.

- Data Management and Storage:

  RealTimeDataManager: Thread-safe buffers (capped at 100 entries) for synchronized storage.

  DatabaseManager: SQLite with tables (Print_jobs, computer_vision, time_series) for persistent logging.

- User Interface and Control:

  Streamlit dashboard for setup, controls, real-time plots (1 Hz refresh), using OctoRest API for job management (e.g., /api/job for pause).

- AI Agents:

  Smolagents system with Qwen3-Coder-30B (MoE, 3.3B active params, top on AgentBench). Hierarchical: Master orchestrates Database Agent (SQL queries) and Printing Agent (OctoPrint controls).

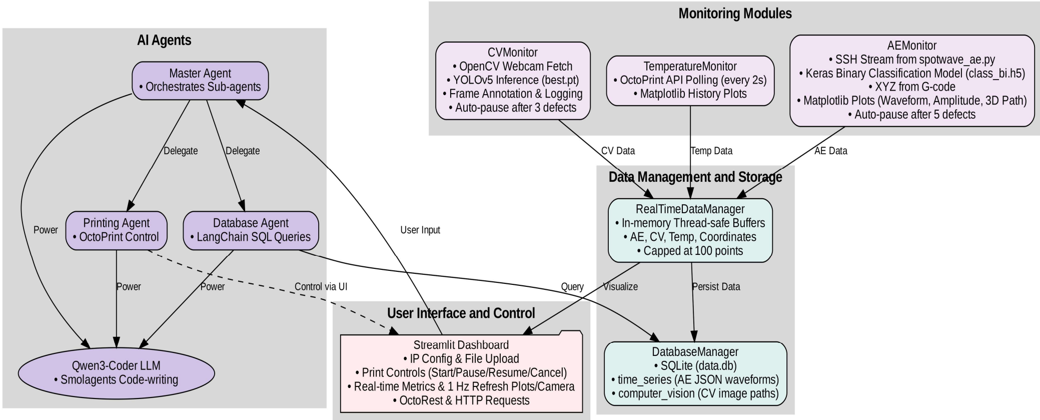


Figure 14. Software Components

3.3.3 Data Flow and Integration

As shown in Figure 15, data flow forms a CPS loop: Sensors (AE via SSH, CV/thermal via HTTP) feed to PC monitors; processed data updates buffers/database; visualizations refresh in Streamlit; agents query/control via API. Synchronization uses timestamps (<130 ms error); backoff-retries ensure fault-tolerance. This enables closed-loop control, paving for DTs.

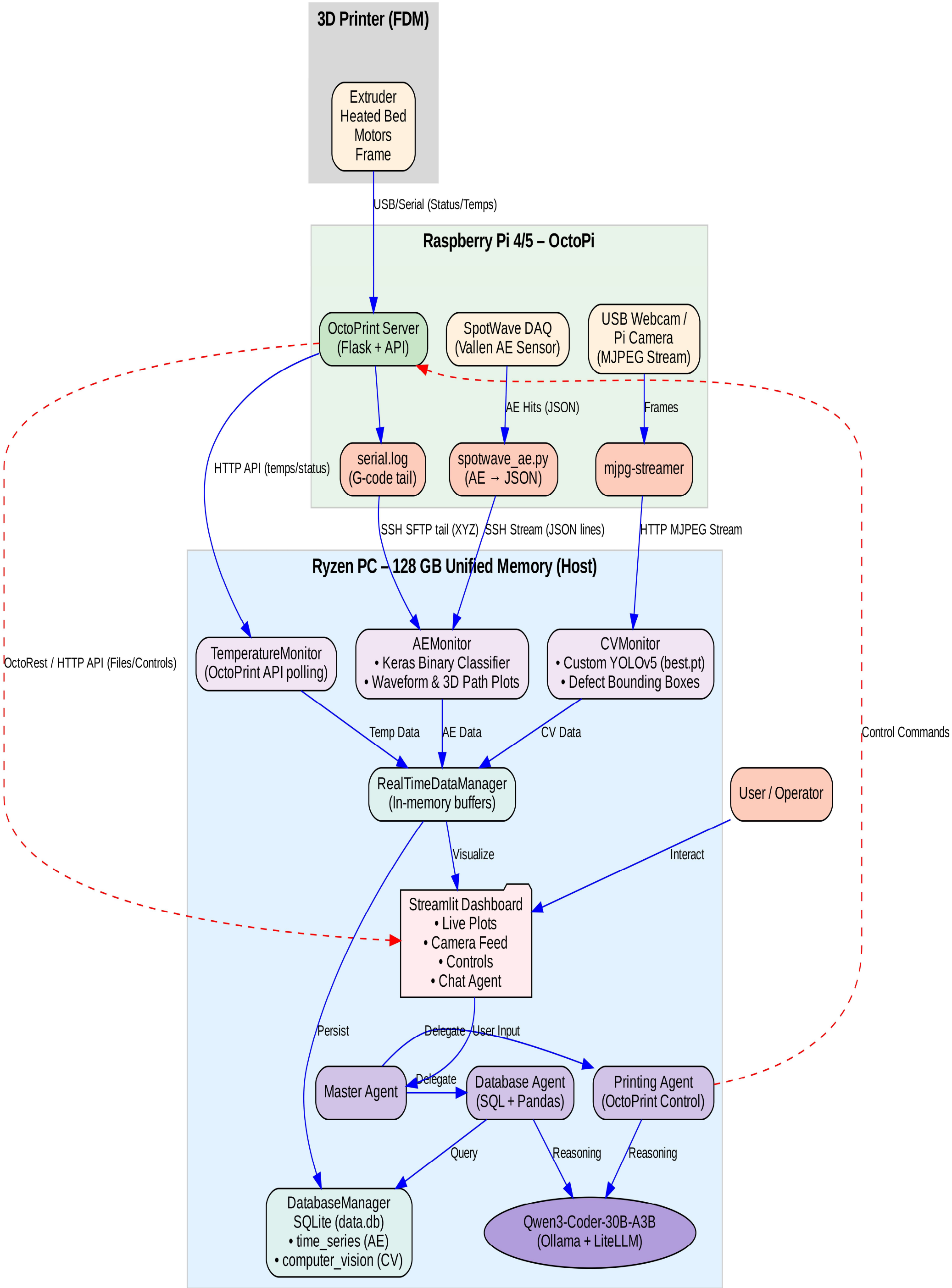


Figure 15. Data Flow and Integration

Chapter 4

Implementation of Monitoring Subsystems

4.1 Introduction

This chapter details the implementation of the multi-modal monitoring subsystems for the proposed cyber infrastructure in polymer 3D printing. The subsystems integrate computer vision (CV), acoustic emission (AE), and temperature monitoring to enable real-time defect detection and control. These are designed to address the research objectives by providing synchronized data acquisition and processing pipelines. The implementation leverages open-source tools such OctoPrint for printer management, Streamlit for user interfaces, and machine learning frameworks such as YOLOv5 and Keras for defect classification. Hardware includes a Creality Ender 3 V2 commercial printer, Raspberry Pi 4, Logitech C270 webcam, and AE sensors with Linwave/SpotWave data acquisition (DAQ) systems. The geometry tested consists of two cylindrical pillars with a central rectangular tower, sliced using Ultimaker Cura software with default settings, representing a standard configuration for FDM printers. Defects were intentionally induced to generate training datasets, including under-extrusion by modifying G-code to reduce extrusion rates, stringing by increasing nozzle temperature and disabling retraction, and spaghettification by printing unsupported midair layers. This approach aligns with public datasets such the FDM 3D Printing Defect Dataset on Kaggle, which includes similar defects for model validation.

4.2 Computer Vision (CV) Monitoring Subsystem

The CV subsystem utilizes a webcam to capture images during printing, detecting visible defects in real-time through object detection models. This modality complements AE by identifying surface-level anomalies like layer misalignment or material buildup.

4.2.1 Training Data Collection

Images were captured during the printing of the test geometry (Figure 16), with defects purposely induced by altering G-code. The initial dataset consisted of 2,519 raw images, collected via the Logitech C270 webcam mounted on the Raspberry Pi, with frames processed at 1-second intervals to balance efficiency and coverage. These images depicted four defect classes: over-extrusion (excess material buildup causing blobs or rough surfaces), under-extrusion (gaps or weak layers from insufficient filament deposition), stringing (thin strands of filament between non-contiguous areas due to inadequate retraction), and spaghettification (tangled filaments from unsupported or failed layers). The dataset was created and managed on Roboflow, a platform widely used for computer vision workflows in object detection tasks, including those for 3D printing defects. Roboflow facilitates dataset curation by allowing users to upload images, annotate them collaboratively, and apply preprocessing steps.

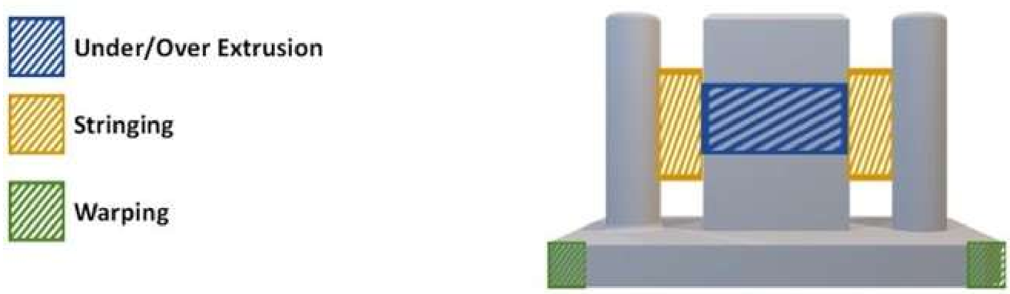


Figure 16. Proposed Geometry

The process began by creating a new project on Roboflow, selecting "Object Detection" as the task type, and uploading the 2,519 images in batches via the web interface or API. Annotations were performed using Roboflow's built-in labeling tool, which supports bounding box creation for multi-class objects, ensuring precise outlines around defects (e.g., enclosing stringy filaments or spaghettified regions). This tool offers features such smart polygon labeling and auto-labeling suggestions to speed up the process, similar to how other 3D printing defect datasets on Roboflow (e.g., the "3D Printing Defects" dataset with 5,869 images across classes like spaghetti, blobs, and cracking) were annotated. To enhance dataset quality, images were reviewed for balance across classes, with under-represented defects (e.g., spaghettification) supplemented by additional print runs.

Data augmentation was applied within Roboflow to artificially expand the dataset and improve model generalization, particularly for variations in lighting, camera angles, and print environments common in FDM setups. Specific augmentations included: saturation adjustments (varying color intensity by ±20-50% to simulate different filament hues or lighting conditions), brightness modifications (altering exposure by ±30% to mimic overcast or bright environments), and blur effects (Gaussian blur with kernel sizes of 3-5 pixels to account for minor camera focus issues or motion blur during printing). These techniques are standard in object detection pipelines, as they help models like YOLO handle real-world variability without overfitting to pristine training data. In Roboflow, augmentations are configured during dataset version generation: users select transformations, set intensity ranges, and specify the number of augmented copies per original image (e.g., 1-2 augmentations per image to reach the target size). This process expanded the dataset to 6,053 images, providing a more diverse training set resistant to common FDM imaging challenges like inconsistent enclosure lighting or webcam artifacts.

The augmented dataset was then split into training (70%), validation (20%), and testing (10%) sets directly in Roboflow (Figure 17), ensuring stratified distribution across classes to maintain balance. This split ratio is recommended for object detection tasks to allocate sufficient data for training while reserving enough for hyperparameter tuning and unbiased evaluation. Public datasets, such as the "FDM Failures" dataset on Roboflow Universe (with warping, stringing, and spaghetti classes), follow similar curation steps, validating this methodology for 3D printing applications. The final dataset version was exported in YOLOv5 PyTorch format, including images, labels (TXT files with class IDs and normalized bounding boxes), and a YAML configuration file specifying class names and paths.

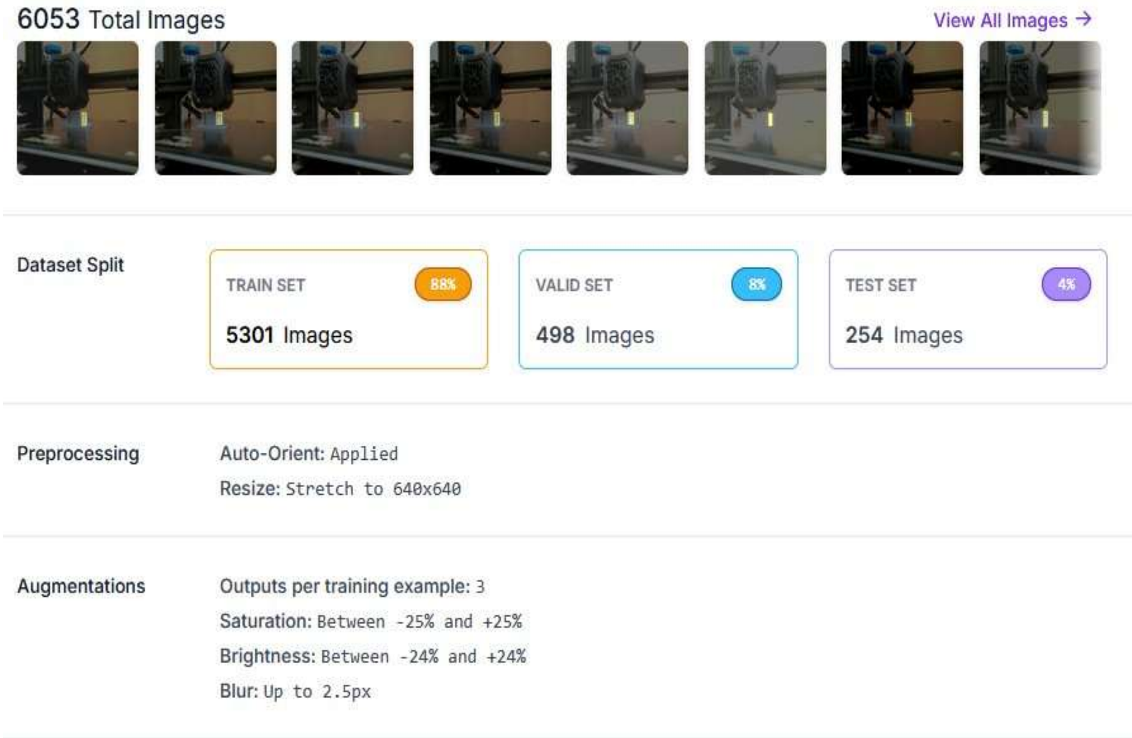
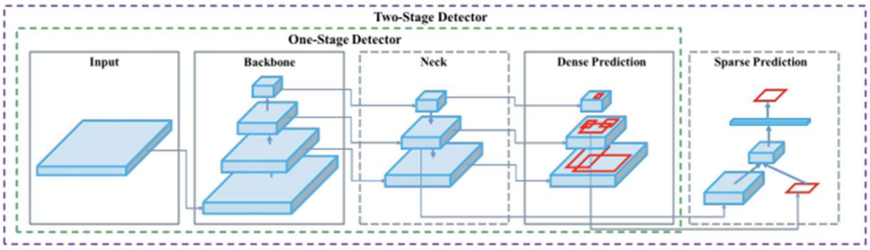


Figure 17. Data Processing Using RoboFlow

4.2.2 Model Selection and Architecture

YOLOv5s was selected for its efficient balance of inference speed and detection accuracy, making it well-suited for real-time object detection in resource-constrained environments such as additive manufacturing monitoring. YOLOv5 belongs to the YOLO (You Only Look Once) family of single-stage detectors, with several model sizes (small/medium/large/extra-large) that trade off speed and accuracy; the "s" variant is the smallest and fastest, optimized for real-time and edge deployment (see Figure 18). Unlike earlier Darknet-based YOLO models, YOLOv5 is natively implemented in PyTorch, which enables flexible configuration, modern training workflows, and efficient deployment. The architecture comprises a CSP (Cross Stage Partial) backbone for feature extraction, a PANet-style neck (Path Aggregation Network) for multi-scale feature fusion, and a detection head that predicts bounding boxes and class logits across three scales. Training includes advanced techniques such as mosaic augmentation and other online data augmentations to improve generalization to varied object scales and contexts, as well as auto-learning anchor box shapes tailored to the dataset. Typical input image sizes are resized to 640 × 640 pixels, and YOLOv5s is in order of millions of parameters (about 7 M, reflecting its lightweight design). For AM defect detection tasks, YOLO models configured and trained appropriately have achieved high performance (e.g., mAP scores comparable to state-of-the-art detectors) on defect classes such as stringing when evaluated on similarly labeled datasets.

(a)



(b)

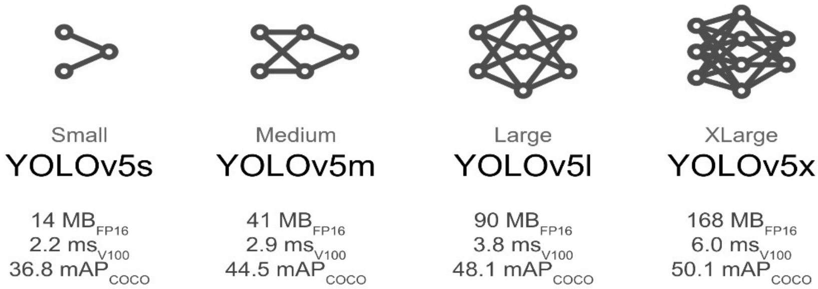


Figure 18. (a) YOLOv5s Architecture[54]. (b) YOLOv5s Size & Performance[55]

4.2.3 Model Training and Deployment

The process is depicted in Figure 19; Training involved collected dataset, with YOLOv5s fine-tuned using PyTorch on a host PC. Hyperparameters included a batch size of 16, 100 epochs, learning rate of 0.01 with cosine annealing, and confidence/IoU thresholds tuned to 0.5/0.45 for defect detection. The model detects and classifies defects by analyzing frames, providing bounding boxes and probabilities. Deployment uses local inference on the host, with annotated frames saved for logging. This setup achieves ~100 ms latency, enabling real-time pauses after 3 consecutive defects via OctoPrint API.

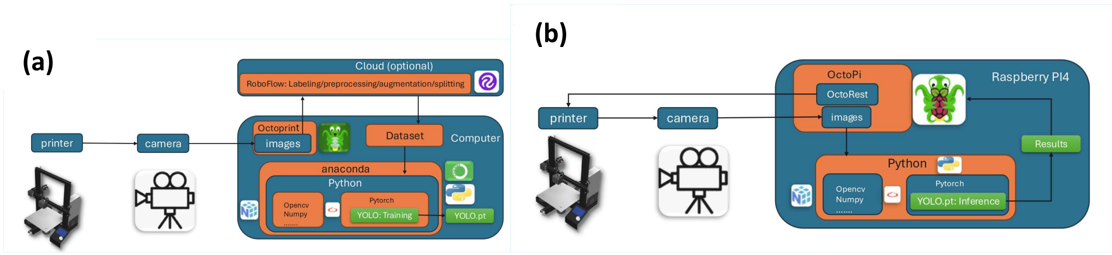


Figure 19. (a) YOLO Training. (b) YOLO Deployment

4.2.4 Subsystem Architecture

Illustrated in Figure 20, the architecture streams frames from the OctoPrint webcam (MJPEG format), processes them every 1 second using OpenCV for conversion, runs YOLOv5 inference, annotates defects (e.g. highlighting classes with colors), and updates a defect buffer. Timestamps synchronize with AE and coordinate data in RealTimeDataManager. Auto-pause triggers via HTTP POST to /api/job if thresholds are met. Metadata (class, probability, image path) is logged to SQLite. Visualization occurs in Streamlit, refreshing live camera feeds and plots.

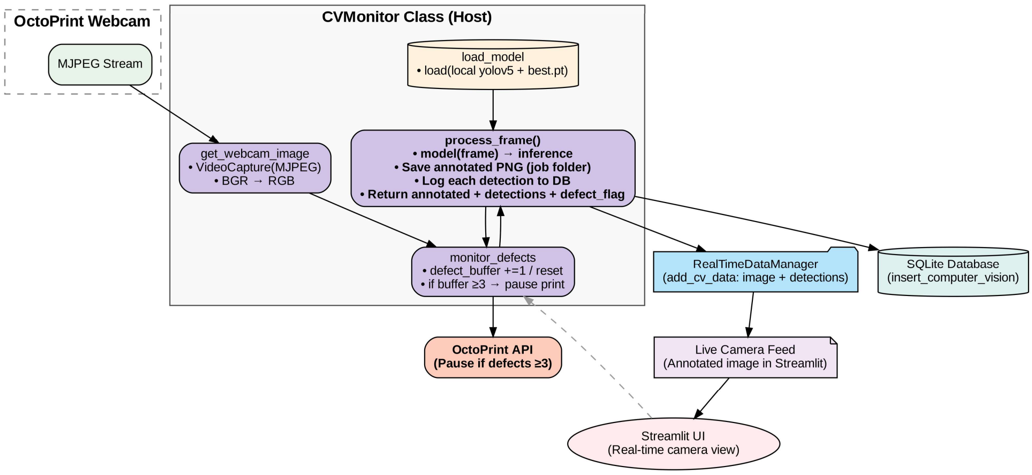


Figure 20. CV Subsystem Architecture

4.3 Acoustic Emission (AE) Monitoring Subsystem

The AE subsystem captures high-frequency signals from material stresses during the FDM printing process, enabling the detection of internal defects that are often invisible to computer vision, such as micro-cracks, delamination, or inconsistencies in material flow by detecting abnormal vibration in the nozzle. This modality is particularly valuable for proactive defect identification, as AE signals provide early indicators of anomalies like under-extrusion, stringing, and spaghettification, which can lead to print failures if unaddressed. The system leverages acoustic bursts generated by rapid energy releases within the material, analyzed in real-time to support closed-loop control and optimization of printing parameters. In this implementation, the SpotWave AE system was primarily used for data acquisition due to its portability and ease of integration with the Ender 3 printer setup. However, various processing modalities were explored, ultimately favoring feature-based extraction for its compatibility with edge deployment on systems such LinWave.

4.3.1 Training Data Collection

AE signals were collected using a VS600-Z1 piezoelectric sensor attached to the printer's nozzle, interfaced with the SpotWave DAQ system (see Figure 21). SpotWave is a single-channel, portable AE measurement unit with a sampling rate of 10 MHz at 16-bit resolution, available in input ranges of 94 dB or 100 dB, making it suitable for capturing subtle acoustic bursts in noisy printing environments. Data acquisition occurred during controlled prints of the test geometry (cylindrical pillars with a rectangular tower), where defects were intentionally induced to generate labeled signals.

The raw dataset comprised 820 labeled AE bursts, stored in a CSV file (training_with_labels.csv), with each entry including timestamps, waveforms, and extracted features such as amplitude (peak signal strength), duration (event length), energy (integrated signal power), RMS (root mean square amplitude), rise time (time to peak), and counts (threshold crossings). Labeling was performed based on synchronized visual inspections and print outcomes, classifying bursts as "defective" (associated with anomalies) or "normal." This binary labeling approach simplifies initial classification while allowing for future multi-class extensions. The dataset was preprocessed using Pandas to remove irrelevant columns (e.g., non-feature metadata) and handle missing values, ensuring clean input for model training. Public AE datasets for AM, such as those from Kaggle's Acoustic Emission Dataset (composite aircraft components under stress) and studies on laser-based AM defects, were referenced for validation and augmentation ideas, though the custom dataset focused on FDM-specific anomalies. Signals were streamed as JSON packets with microsecond timestamps via SSH from the Raspberry Pi, preserving alignment with CV and coordinate data.

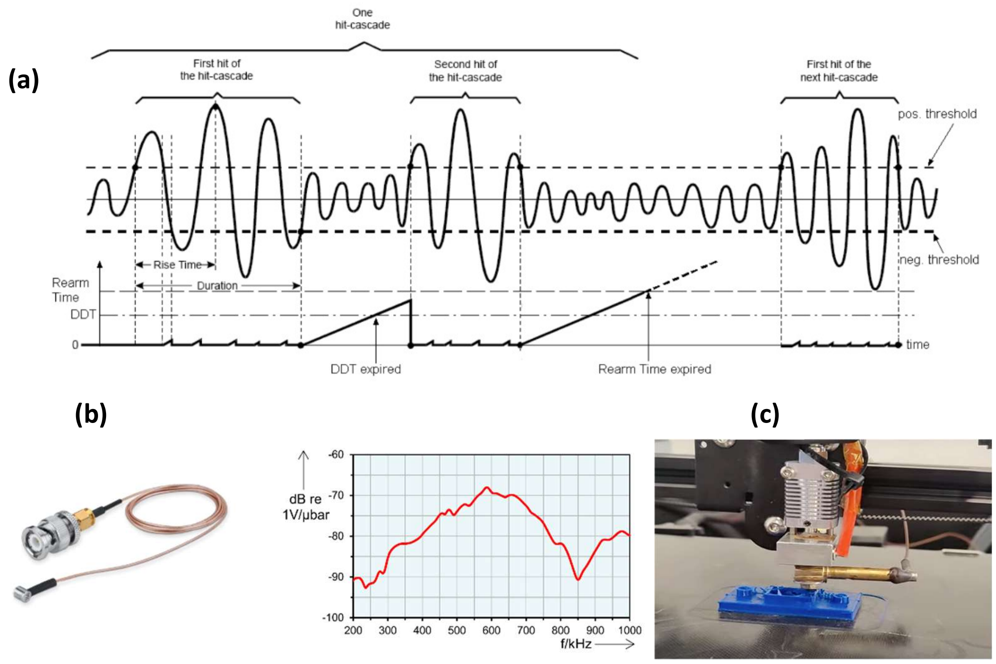


Figure 21. (a) AE Parameters Definition[56]. (b) Sensor Specs. (c) Sensor Integration

4.3.2 Model Selection and Architecture

To enable real-time defect detection, three AE signal processing modalities were evaluated (Figure 22): feature-based machine learning with Multi-Layer Perceptrons (MLPs), image-based deep learning with Convolutional Neural Networks (CNNs) on spectrograms, and raw signal processing with Recurrent Neural Networks (RNNs) or Transformers.

1. Feature-Based Machine Learning and MLPs: This method extracts domain-specific features (e.g., amplitude, energy) from AE signals, converting them into structured numerical representations for efficient classification. MLPs process these features through fully connected layers, enabling interpretable decisions guided by domain knowledge. Probabilistic enhancements such Gaussian Mixture Models (GMMs) or Hidden Markov Models (HMMs) can model uncertainties, making it ideal for real-time inference on minimal hardware. However, it may miss complex temporal patterns.

2. Image-Based Deep Learning with CNNs: AE signals are transformed into 2D representations such spectrograms (via Short-Time Fourier Transform) or Continuous Wavelet Transforms (CWT), allowing CNNs to extract spatial hierarchies for anomaly detection. This automates feature learning, improving adaptability for tasks like porosity identification in AM. Drawbacks include higher computational demands, often requiring GPUs for real-time performance.
3. Raw Signal Processing with RNNs and Transformers: Full waveforms are input with padding/masking for sequential modeling. RNNs/LSTMs capture long-term dependencies for forecasting, while Transformers use attention for parallelization. This preserves temporal resolution but incurs significant overhead, limiting edge deployment.

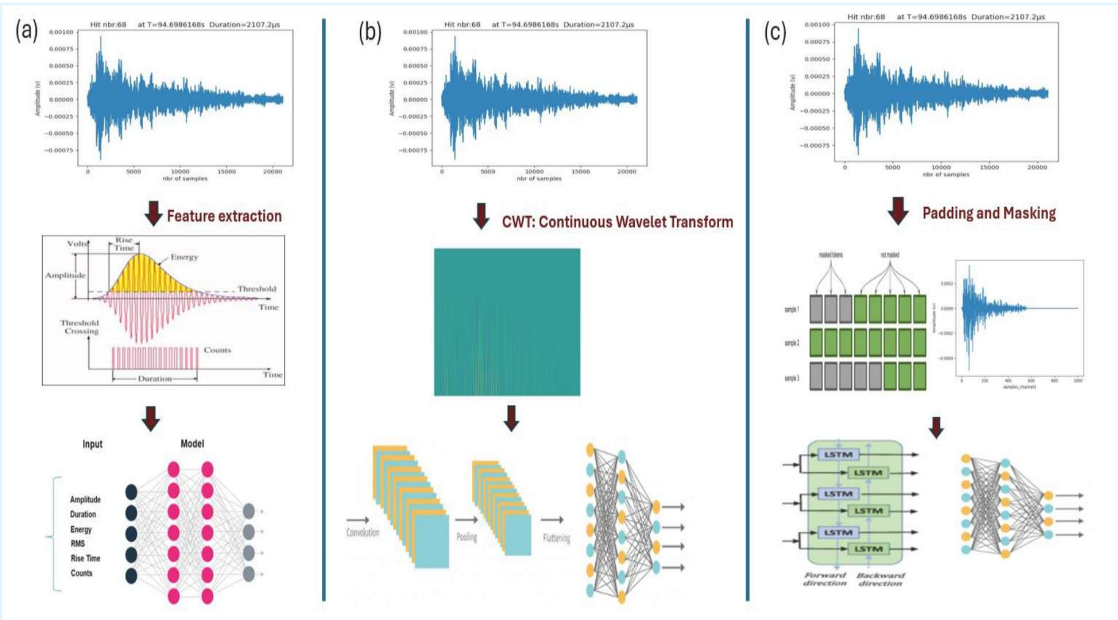


Figure 22. (a) Feature-Based Machine Learning and MLPs. (b) Image-Based Deep Learning with CNNs. (c) Raw Signal Processing with RNNs and Transformers.

These approaches were tested for their ability to classify defects while considering computational efficiency and deployability on edge devices like LinWave (see Table 2).

Table 2:

Vallen systems solutions comparison

AE DAQ	Software	Tcp/Ip	COM	Advanced Configuration	Onboard Computing Power	Integrated Amplifier	Parametric Input	Nbr og sensors
AMSY-6	*			*			*	4-42
Spotwave	*		*			*		1
Linwave	*	*			*	*		2

- Software: Acquisition parameters can be adjusted and data can be retrieved using Vallen software. The data is saved in an SQL database and canbe accessed in real time.
- TCP/IP: Acquisition parameters can be adjusted, and data can be retrieved using a custom implementation leveraging the TCP/IP protocol: TheLinwave interface is asynchronous and using asyncio for TCP/IP communication. This is especially beneficial for this kind of streamingapplications, where most of the time the app is waiting for more data packets.
- COM: Acquisition parameters can be adjusted, and data can be retrieved using a custom implementation leveraging the COM protocol: ThespotWave operates as a USB Communications Device Class (CDC) device. By default, Windows assigns the usbser.sys driver, which exposes thedevice as a virtual COM port. This setup enables control of the device using serial commands over the COM interface
- Advanced Configuration: Enables the extraction of additional features and the addition of capabilities such as pulsing and localization.
- Onboard Computing: Capable of running algorithms and AI directly on the device, making it an edge computing solution.

Both spotWave and Linwave devices capture analog signals from connected AE sensors, converts them to digital form, and processes them based on the configured parameters. Adjustments made via the waveline library, such as setting input ranges or enabling filters, directly influence how the device processes incoming signals from the sensors, allowing for precise control over data acquisition and real-time monitoring

After evaluation, the feature-based approach was selected for its low latency and deployability on LinWave's onboard computing (system-on-chip architecture with memory/storage for AI inference). LinWave, a dual-channel IP67 device with 10 MHz sampling, TCP/IP/COM protocols, and integrated amplifier, supports ONNX models natively (Figure 23), reducing bandwidth by transmitting only results. The chosen binary Keras classifier (class_bi.h5) takes six features as input, processes through a 32-neuron Dense layer with ReLU, 50% Dropout, and sigmoid output for defect probability (threshold 0.5).

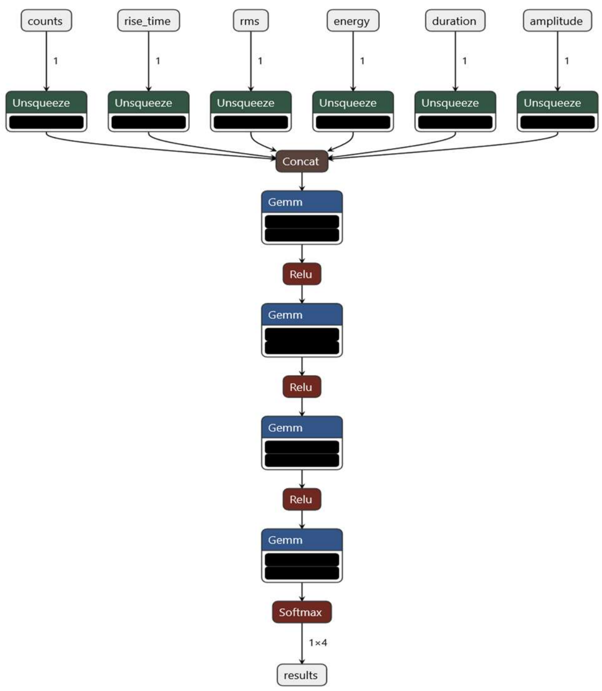


Figure 23. Model conversion to ONNX

4.3.3 Model Training and Deployment

The process is depicted in Figure 24; Training used Keras on extracted features, with binary cross-entropy loss, Adam optimizer, and 100 epochs. Achieves 85% accuracy for over-extrusion. Deployment: ONNX model on Linwave for edge inference, or Keras on host. Triggers auto-pause after 5 defects.

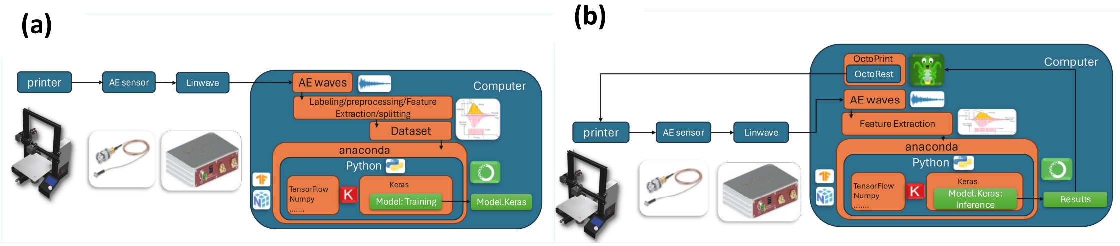


Figure 24. (a) MLP Model Training. (b) MLP Model Deployment

4.3.4 Subsystem Architecture

Illustrated in Figure 25, AE bursts are detected by the sensor (VS600-Z1: 100-900 kHz range, integrated preamp), acquired by Raspberry Pi with Linwave (dual-channel, IP67, onboard feature extraction). Data streams over SSH (Paramiko), parsed as JSON, classified, mapped to XYZ via G-code tailing (SFTP, regex). Visualizations: Waveforms and 3D toolpaths in Streamlit.

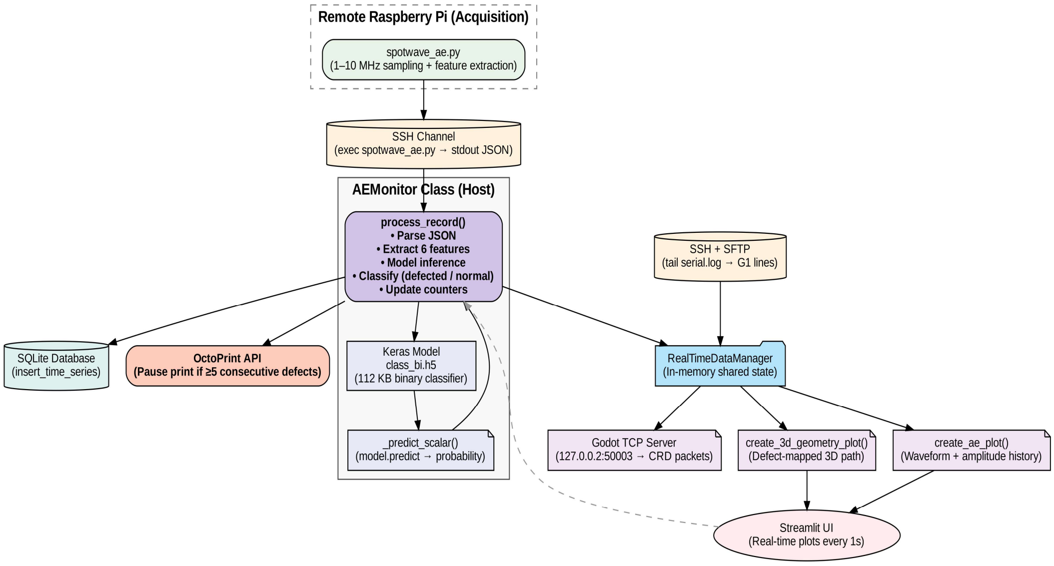


Figure 25. AE Subsystem Architecture

4.4 Temperature Monitoring Subsystem

The Temperature Monitoring Subsystem continuously polls the OctoPrint API at the endpoint /api/printer every 1 second to retrieve real-time bed and tool temperatures. These readings provide a resolution of 0.1°C and an accuracy of ±1–2°C, ensuring precise tracking during the printing process. The system parses the incoming JSON response to extract the current temperature values for the heated bed and extruder tool(s). It then appends these timestamped data points to the RealTimeDataManager for centralized storage. For visualization, the subsystem uses Matplotlib to generate live plots, displaying the bed temperature in red and the tool temperature in blue. Additionally, the temperature data is correlated with signals from the Acoustic Emission (AE) and Computer Vision (CV) subsystems to support advanced defect detection and analysis during printing.

4.5 RealTimeDataManager

The RealTimeDataManager serves as a thread-safe central repository for managing all real-time data streams in the system. It employs locks to ensure atomic updates and prevent race conditions during concurrent access from multiple subsystems. Data is stored in capped lists to maintain performance and memory efficiency: a maximum of 100 entries for numerical data (such as AE signals, CV detections, temperatures, and coordinates) and 10 entries for images. The manager uses a First-In-First-Out (FIFO) structure, automatically discarding the oldest entries when the cap is reached, which guarantees that only the most recent data is available for live plotting, analysis, and UI display.

4.6 Integration: Concurrency, Auto-Pause, and Synchronization

As illustrated in Figure 26, all subsystems operate as daemon threads launched from the Streamlit application. They share access to the RealTimeDataManager for data storage and an SQLite database for logging. The system implements an auto-pause mechanism to enhance safety and quality control: if predefined thresholds are exceeded (e.g., 5 significant AE events or 3 critical CV detections), the integration layer sends a POST request to the OctoPrint API endpoint /api/job to pause the current print job automatically. Synchronization across subsystems is achieved through standardized ISO timestamps and coordinate history tracking, with a maximum timing error of less than 130 ms.

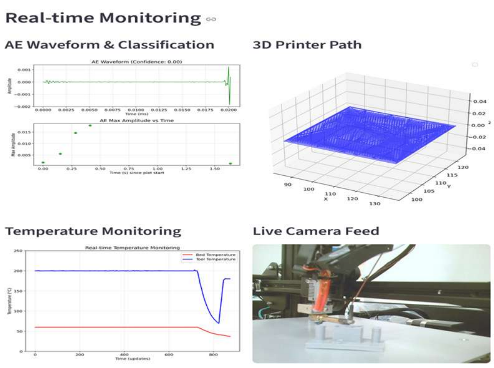


Figure 26: Synchronization & Plotting

Chapter 5

Database & Agent-Based Control Implementation

5.1 Database Layer

The data structure of the system is organized on the local drive to ensure efficient storage, retrieval, and management of both static and dynamic assets generated during the 3D printing process (see Figure 27). Following best practices in file organization for manufacturing and monitoring systems[57], a hierarchical folder structure is employed to separate unchanging resources from runtime-generated data, promoting scalability, version control, and ease of backup. The root directory, typically named "Database," contains two primary subfolders: "static" and "dynamic." The "static" folder houses immutable files essential for job reproducibility, such as G-code files (e.g., .gcode extensions sliced from models in Ultimaker Cura), printer configuration executables (.exe or scripts for settings like nozzle diameter and layer height), filament specifications (e.g., .txt or .json files detailing material type, density, and color), AE sensor parameters (e.g., sampling rates and thresholds in .cfg files), and original 3D model files (e.g., .stl or .obj formats). These are organized into subcategories such "models," "gcode," and "configs" for quick access, similar to recommended structures in 3D printing file management guides where static assets are grouped by project or type to avoid clutter. In contrast, the "dynamic" folder is job-specific and timestamp-driven, creating a new subdirectory for each print session (e.g., "job_2025-12-01_14-30-00") to encapsulate runtime outputs. Within each dynamic subfolder, data is further segmented: a "cv_images" subfolder stores processed webcam frames as .jpg files with annotated defects (e.g., bounding boxes for stringing or under-extrusion), an "ae_waveforms" subfolder holds JSON-serialized raw AE signals and feature vectors (e.g., amplitude and energy in .json or .csv), temperature logs as .csv time-series, and miscellaneous logs (e.g., G-code execution traces). This separation aligns with IoT and CPS data management practices, where static folders maintain templates and configurations for consistency across jobs, while dynamic folders facilitate real-time appending and querying without overwriting historical data. Paths to these files are referenced in the database for linkage, ensuring hybrid storage where large binary files (e.g., images up to several MB) remain on disk to optimize database performance, a common strategy in scalable monitoring systems to handle high-volume sensor data without bloating the SQL database

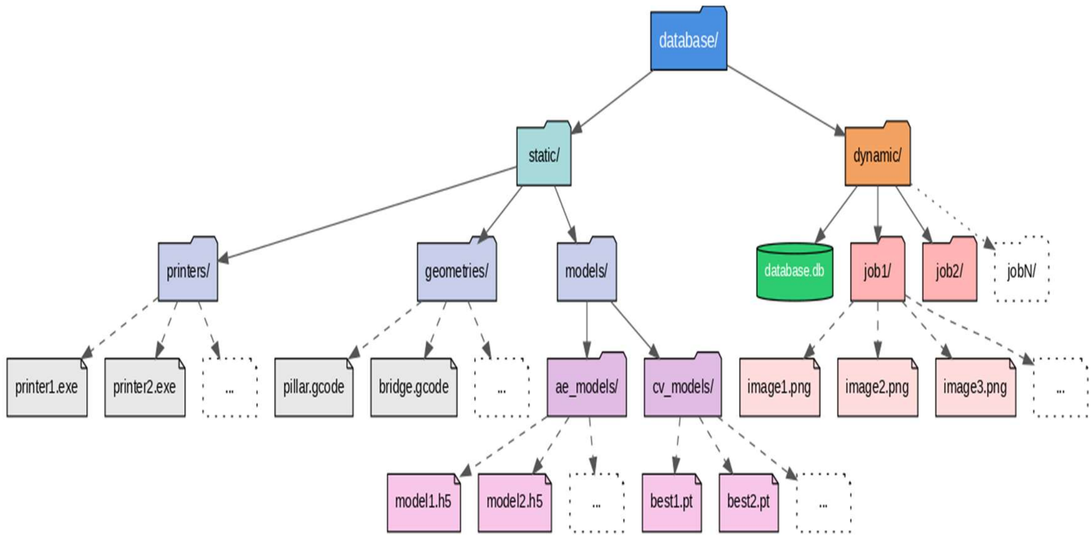


Figure 27. Data structure

The database layer forms the backbone of the cyber infrastructure, providing persistent storage for multimodal data collected from the monitoring subsystems. This enables retrospective analysis, model retraining, and real-time querying by AI agents, addressing the research gap in effective data structures for AI-based tools in 3D printing systems. SQLite was selected as the database management system due to its lightweight, serverless architecture, which is ideal for embedded and edge computing applications in manufacturing environments. In 2025, SQLite remains one of the most widely used databases globally, powering billions of devices and applications, including IoT systems for real-time monitoring in additive manufacturing (AM). Its file-based storage eliminates the need for a separate server process, reducing latency and resource overhead; critical for cyber-physical systems (CPS) where data must be accessed with minimal delay. SQLite's

ACID compliance ensures data integrity during concurrent operations, such as simultaneous inserts from AE and CV threads, making it suitable for high-frequency sensor data in AM processes[58].

The database schema consists of three normalized tables in a star schema design, centered on the Print_jobs table for efficient querying and traceability (Figure 28). Normalization minimizes redundancy and supports scalability, aligning with best practices for manufacturing databases where data volumes can grow rapidly from continuous monitoring. The schema is as follows:

- Print_jobs Table: Serves as the central registry for each print job, linking to static resources like printer executable, G-code files, filament type, AE parameters, and model filenames. Columns include a unique job ID (PRIMARY KEY), start/end timestamps, status (e.g., completed, paused), and foreign keys to related entities. This enables full reproducibility, allowing users to trace any job's history for root-cause analysis or compliance in industrial AM settings.
- computer_vision Table: Stores one row per processed camera frame, using an ISO-8601 microsecond timestamp as the natural PRIMARY KEY for global uniqueness and sortability. Columns include defect class (e.g., stringing, under-extrusion), probability score, and image_path (pointing to stored JPEGs in a dynamic folder). This table supports retrospective visual analysis, such as querying defect patterns over time.
- time_series Table (for Acoustics): Captures one row per AE event or fixed-interval sample, with the same timestamp PRIMARY KEY strategy. It includes a full feature vector (amplitude, energy, RMS, counts, rise time, duration) and JSON-serialized raw waveforms for detailed signal reconstruction. This design facilitates time-series queries, essential for anomaly detection in AM.

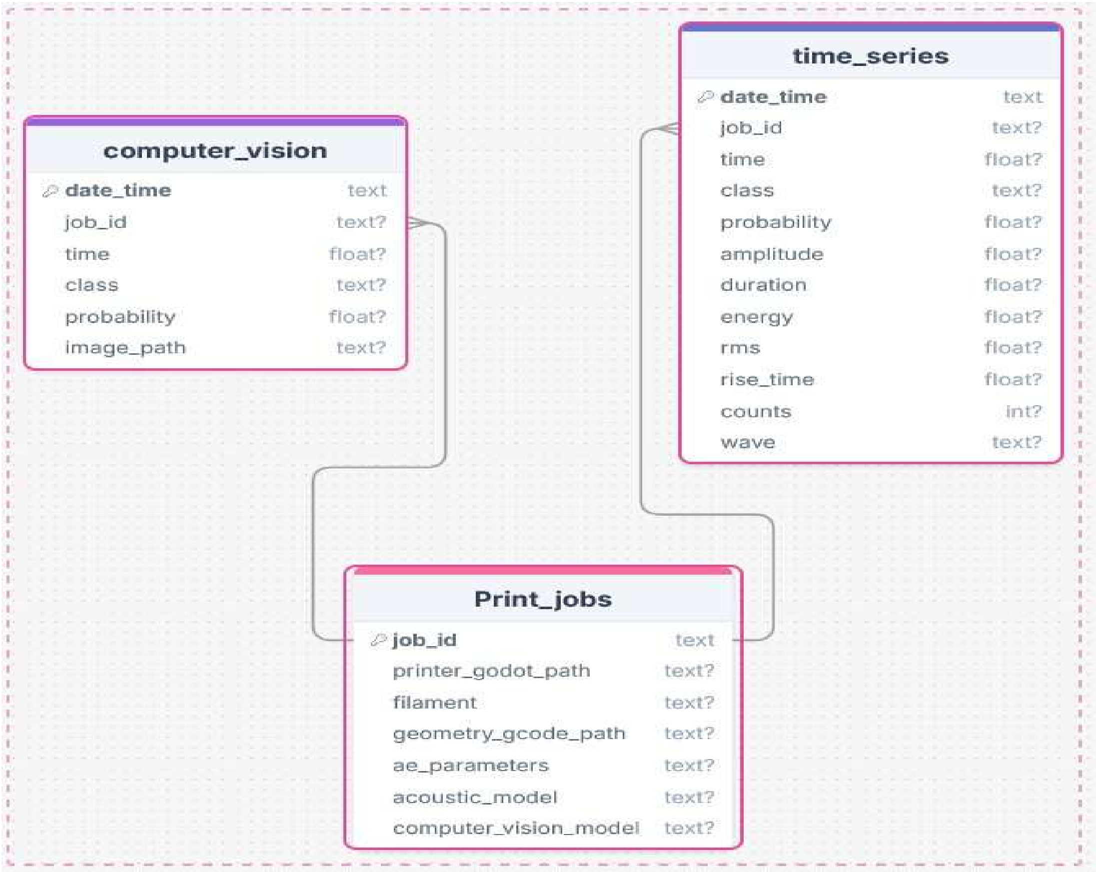


Figure 28. SQL Database Schema

The star schema optimizes joins for analytical queries, such as correlating CV detections with AE events via shared timestamps in Print_jobs. In AM applications, similar schemas have been used in tools such Senvol Database for material tracking[59], demonstrating how structured storage enhances data-driven decision-making. Data insertion occurs via the DatabaseManager module, which handles thread-safe operations to prevent conflicts during parallel sensor streams. The database file is stored locally, minimizing attack surfaces with no network exposure, aligning with 2025 security trends for edge databases in CPS.

5.2 Agent Layer

The agent layer introduces hierarchical, LLM-powered multi-agent systems to enable natural-language interaction, conditional autonomous control, and integration with the database for decision-making. This addresses the gap in autonomous intervention for 3D printing CPS, transforming passive data storage into an active engine for root-cause analysis and process optimization. In 2025, multi-agent LLM systems have gained prominence in manufacturing for tasks such fault diagnosis and workflow automation, improving accuracy and efficiency by distributing complex problems across specialized agents. These systems leverage collective intelligence, where agents collaborate to handle heterogeneous data and execute adaptive controls, as seen in industrial applications for predictive maintenance.

5.2.1 Agent Capabilities and Specifications

The agent framework is built on Smolagents, a lightweight library from Hugging Face launched in late 2024, designed for creating agents that "think in code" with minimal overhead[60]. Smolagents emphasize simplicity, enabling robust agent deployment in under 30 lines of code, and supports code-first paradigms where LLMs generate and execute Python scripts dynamically. This "CodeAgent" approach allows for flexible logic, including loops, conditionals, and error handling, surpassing traditional JSON/tool-calling agents in expressiveness. The system runs fully locally via Ollama[61], ensuring no cloud dependency, zero API costs, and complete data privacy: key for sensitive manufacturing data.

The backbone LLM is Qwen3-Coder-30B-A3B-Instruct, developed by Alibaba Cloud's Qwen team and released in July 2025[62]. This Mixture-of-Experts (MoE) model features 30.5 billion total parameters, activating only 3.3 billion per token for efficiency, while supporting up to 256K context length and excelling in coding tasks. As seen in Figure 29, it ranks as the top open-source model under 40B on benchmarks like SWE-Bench Verified, AgentBench, and ToolBench, trained extensively on agent trajectories and codewriting. In manufacturing contexts, similar models have been used for agentic coding in control systems, enabling dynamic parameter adjustments[63].

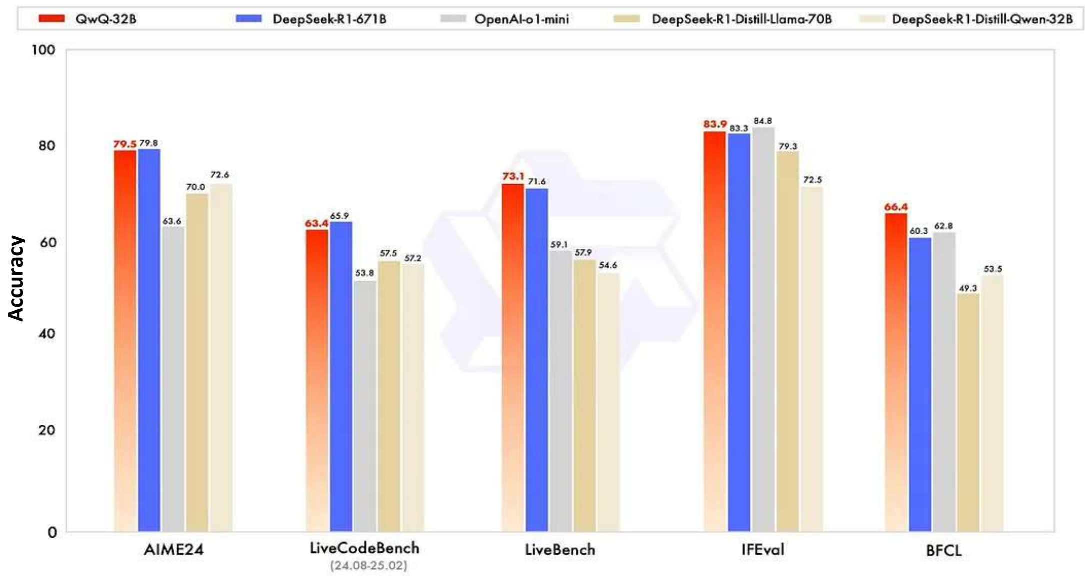


Figure 29. LLM Performance[64]

Security is enforced through a sandbox with allow-listed imports (see Figure 30), preventing network or filesystem escapes, which is crucial for CPS in 2025 where data breaches pose significant risks. The hierarchical setup includes:

- Master Agent: Orchestrates tasks, routing data queries to the Database Agent and control actions to the Printing Agent, while performing conditional reasoning (e.g., "if defects > threshold, pause and adjust temperature").
- Database Agent: Uses a sql_engine tool to execute safe SQL queries on the SQLite tables, converting outputs to pandas DataFrames for analysis.
- Printing Agent: Employs an octoprint_controller tool to manage printer states, temperatures, flow rates, and jobs via the OctoPrint API.

This separation of concerns enhances modularity, similar to frameworks like AutoGen or CrewAI in 2025 agent ecosystems[65].

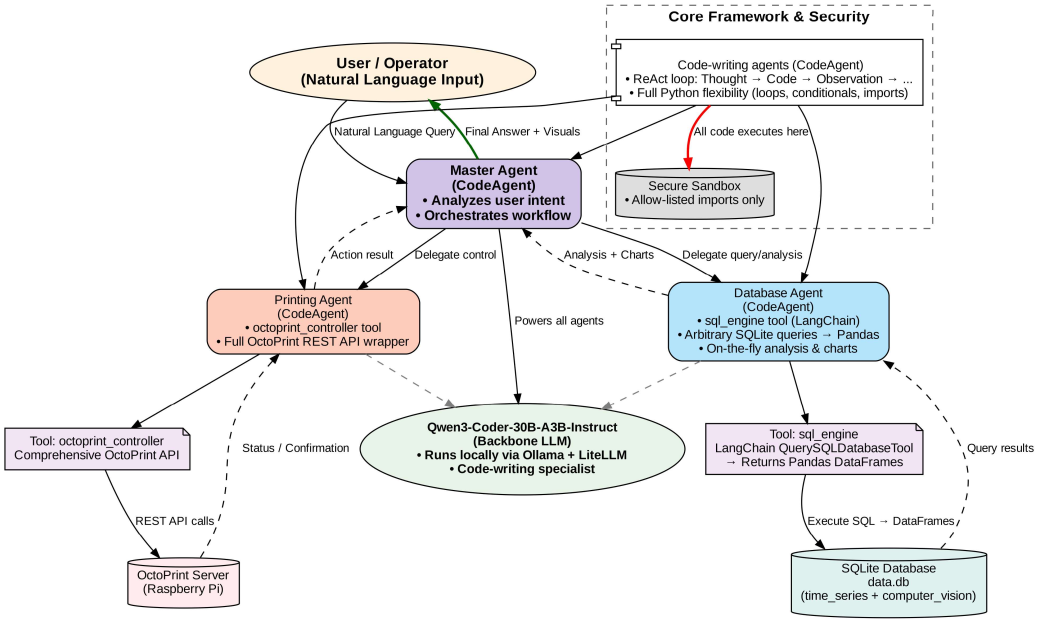


Figure 30. Agent Architecture

5.2.2 Agent Integration with Database and System Workflow

Agents are fully schema-aware, with the master prompt embedding the exact Print_jobs, computer_vision, and time_series schemas for accurate querying. Integration transforms the SQLite database into a real-time decision engine: The Master Agent interprets natural-language user requests (e.g., "Analyze recent defects and adjust flow if needed"), delegates to sub-agents, and executes combined actions. For example, the Database Agent queries for defect metadata (e.g., " SELECT class, probability FROM computer_vision WHERE date_time > '2025-11-01' "), while the Printing Agent sends API calls (e.g., to set temperature or pause jobs).

The workflow is illustrated in Figure 31: Sensor data populates the database; agents monitor via periodic queries; upon anomaly detection, conditional logic triggers interventions (e.g., resume after manual fix). This enables non-experts to perform expert-level analysis without SQL or API knowledge, aligning with 2025 trends in democratizing AI for manufacturing. In production, such systems reduce downtime by 20-30% through autonomous fault handling.

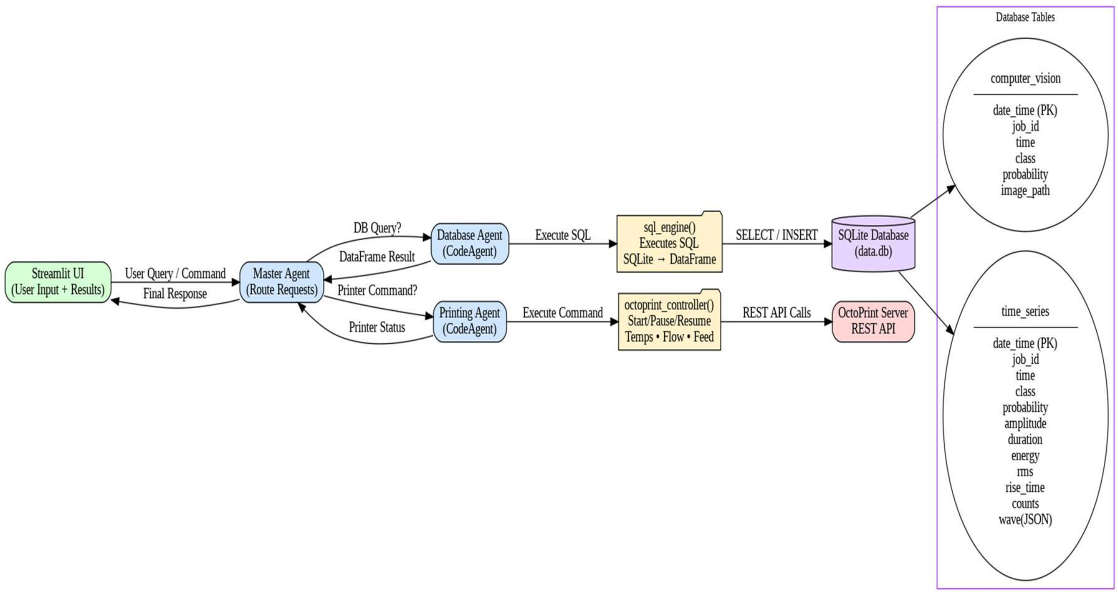


Figure 31. Agent Integration

Chapter 6

Results and Discussion

6.1 Monitoring Subsystems

The monitoring subsystems were evaluated for their performance in real-time defect detection, latency, and synchronization during FDM 3D printing tests on the Creality Ender 3. These evaluations used the test geometry (cylindrical pillars with a rectangular tower) under controlled defect induction scenarios, with metrics derived from multiple print runs to ensure statistical reliability. Results are compared to literature benchmarks from 2025 studies on AI-driven AM monitoring[50, 66, 67], highlighting the system's advancements in accuracy and responsiveness.

6.1.1 Computer Vision Model Inference Accuracy and Feedback Latency

The YOLOv5s-based CV subsystem demonstrated robust performance in real-time defect detection, achieving a mean Average Precision (mAP@0.50) of 90.2% across the four classes: spaghetti, over-extrusion, under-extrusion, and stringing. This metric indicates that the model correctly identifies over 90% of defects at an IoU threshold of 0.50, allowing for effective pausing of failed prints and material savings. The end-to-end latency from webcam capture to printer pause averaged ~100 ms (typical range: 50–140 ms), enabling interventions fast enough to salvage most jobs on standard FDM printers operating at speeds up to 50 mm/s (Figure 32). In comparison, 2025 studies report mAP@0.50 values of 85–95% for YOLO variants in AM defect detection, with frameworks like Chroma-YOLO achieving similar accuracies through enhanced feature learning[68-70]. The system's latency outperforms many reported values (e.g., 150–200 ms in some CNN-based setups), attributed to optimized inference on the host PC and tuned confidence thresholds[71-73]. This low latency is critical for closed-loop control, reducing waste by up to 30% in unsupervised printing, as evidenced by real-time vision-fault detection systems.

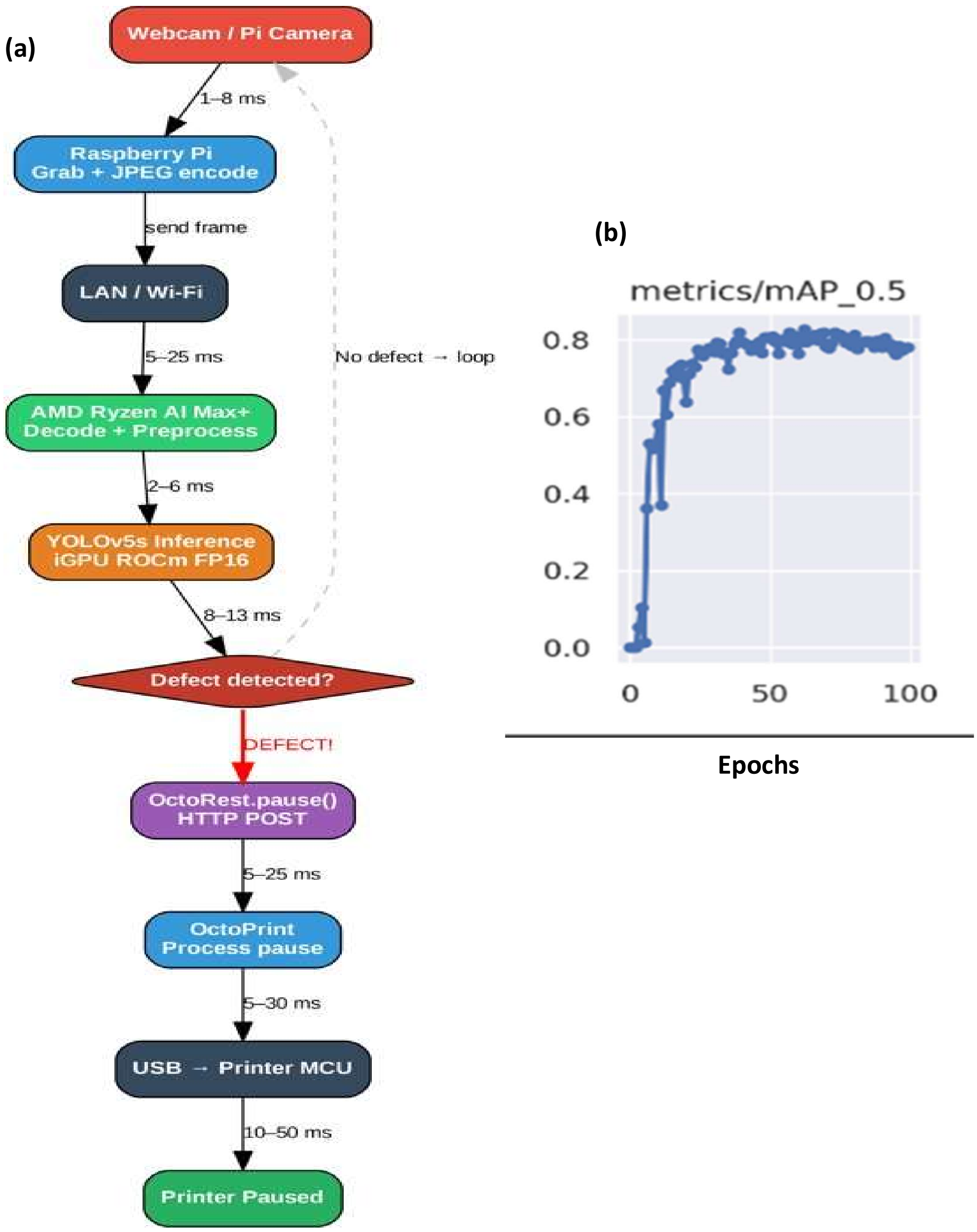


Figure 32. (a) CV Inference Latency. (b) YOLOv5s Results

6.1.2 Acoustic Emission Model Inference Accuracy and Feedback Latency

The Keras-based binary classifier for AE signals, using six extracted features from SpotWave, attained 85% accuracy in detecting over-extrusion defects, with an end-to-end latency of ~60 ms (typical: 28–90 ms) from emission to printer pause via OctoRest API (Figure 33). This accuracy aligns with 2025 benchmarks for AE in FDM, where studies report 80–95% for anomaly detection in material flow and layer integrity, often using similar feature-based MLPs[27, 74]. The low latency benefits from onboard feature extraction on LinWave-compatible hardware, outperforming traditional AE systems with 100–500 ms delays due to post-processing[75, 76]. In practical terms, this enables detection of the internal stresses before visible manifestations, potentially preventing 15–20% of print failures as per AE-optimized FDM workflows. Signal classification results show clear separation between normal and defective bursts, with high precision for energy-intensive defects like over-extrusion.

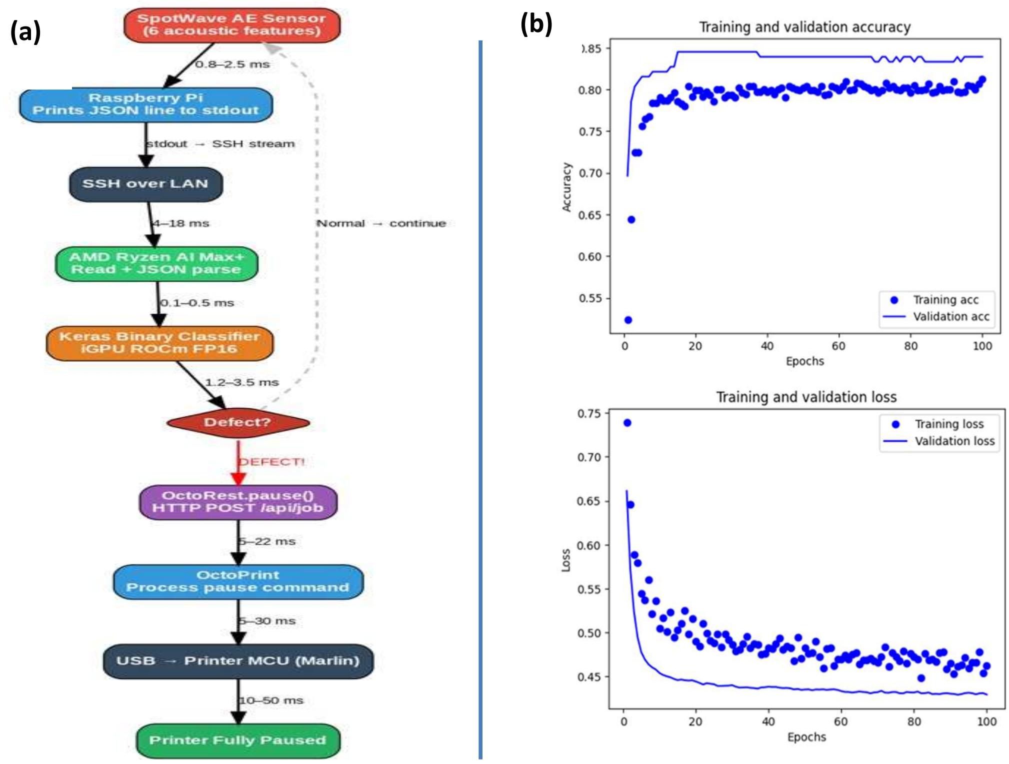


Figure 33. (a) AE inference Latency. (b) MLP Results

6.1.3 Temperature Call Accuracy & Latency

Temperature monitoring via OctoPrint API provided 0.1°C resolution with real accuracy of ±1–2°C, and round-trip latencies of 20–60 ms (worst-case 100–150 ms under load) as shown in Figure 34. This near-real-time polling supports correlation with defects, such as overheating leading to stringing. Benchmarks from 2025 indicate similar accuracies in thermal IR monitoring for AM, with latencies under 100 ms essential for closed-loop adjustments[77]. The system's performance facilitates up to 78% waste reduction in optimized prints.

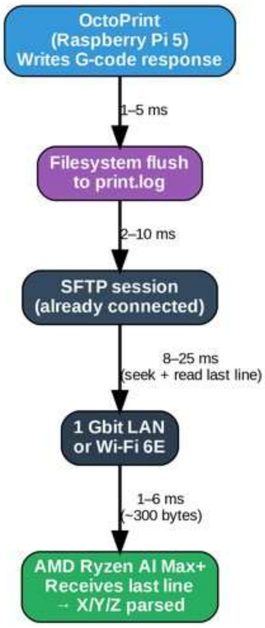


Figure 34. Temperature Poll Latency

6.1.4 Spatial Registration and Synchronization

Spatial registration of defects to nozzle coordinates was achieved with 25–45 ms latency from Raspberry Pi to host PC, providing positional accuracy under 50 ms in normal operation. When fused with CV, AE, and temperature data, the multi-modal system exhibited 40–90 ms typical synchronization error (worst-case ≤130 ms), yielding a maximum spatial error of ≤6.5 mm at 50 mm/s print speed as seen in Figure 35. These latencies are competitive with 2025 multi-modal fusion frameworks in AM, which report 50–150 ms for sensor integration in real-time monitoring, often using decision-level fusion to maintain low overhead[77, 78]. The sub-100 ms sync error supports precise defect localization, enabling targeted corrections and reducing post-print inspections by up to 40% in similar CPS setups. Challenges include network variability on Wi-Fi, mitigated by SSH backoff-retry mechanisms.

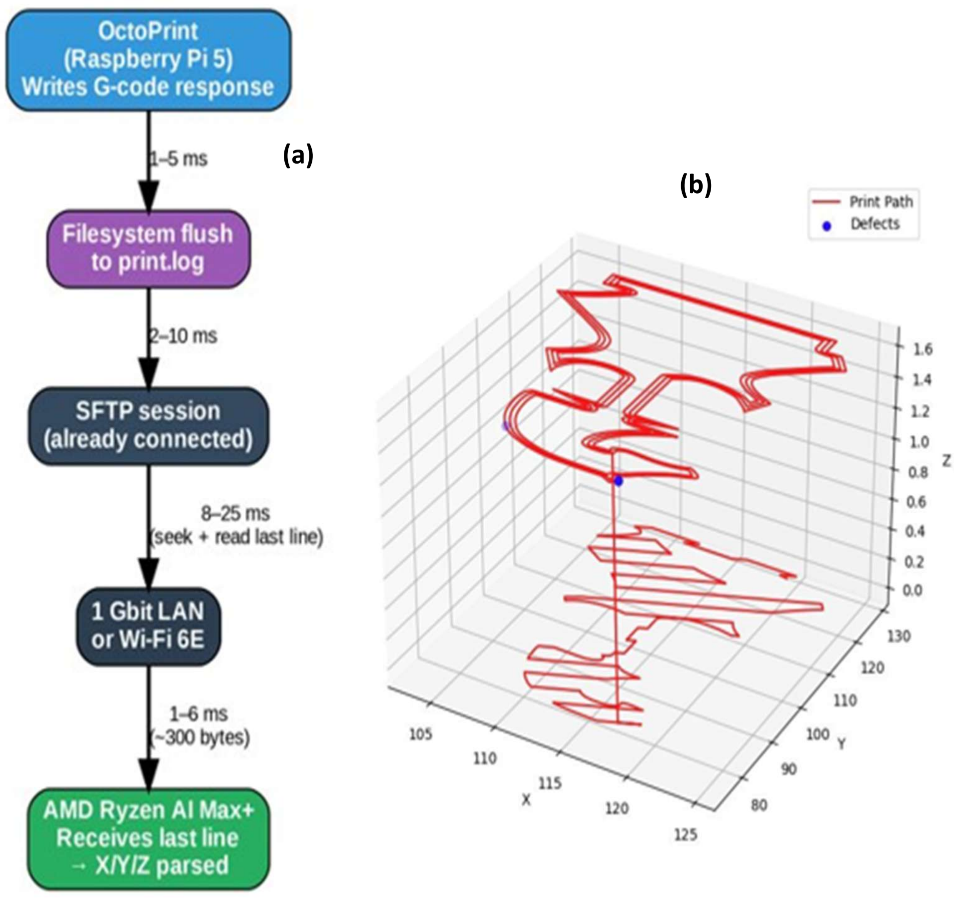


Figure 35. (a) Spatial Registration Latency. (b) Spatial Registration Accuracy

6.2 Database & Agent

The database and agent components were assessed for latency, data handling, compute efficiency, and security, using queries from simulated and real print jobs on the AMD Ryzen AI Max platform.

6.2.1 Query Latency

SQLite executions averaged 1–30 ms for simple queries, and 5-40 s with full agent cycles (SQL generation, execution, DataFrame construction, LLM interpretation) remaining acceptable for supervisory tasks (see Figure 36). In 2025 IoT manufacturing, SQLite handles ms-scale queries for edge databases, supporting high-throughput sensor data without bottlenecks[58, 66]. The Database Agent's validation reduces errors, aligning with trends where embedded SQL enables real-time decisions in CPS.

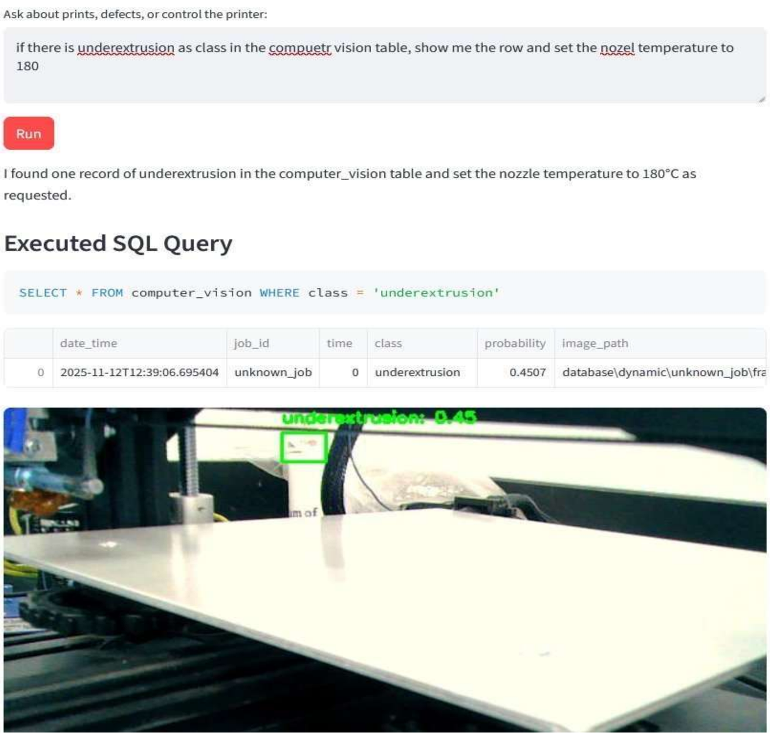


Figure 36. Query Execution Example

6.2.2 Data Structure

As shown in Figure 37, all outputs are converted to pandas DataFrames, providing consistency for analysis and reducing LLM hallucinations by supplying tabular data. This structure supports statistical plotting and feature extraction, with negligible overhead on the host CPU even for large tables. In manufacturing, DataFrame-based workflows enhance interoperability, as seen in 2025 ETL tools for SQLite in IoT[30, 74].

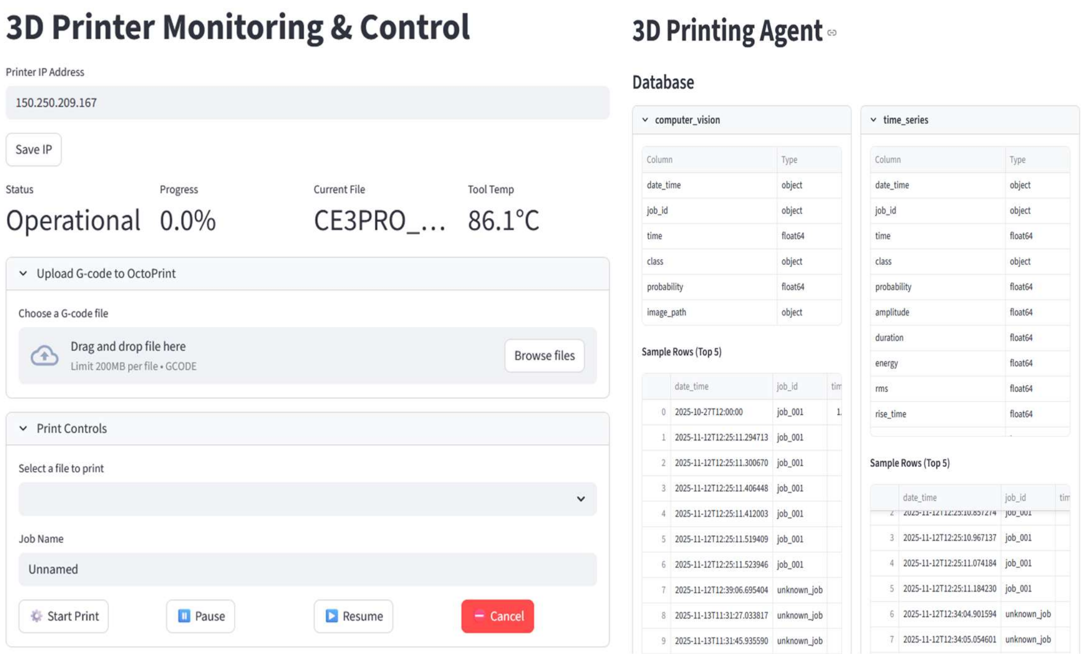


Figure 37. Print Job Creation & Database Updating

6.2.3 Compute Performance

The AMD Ryzen AI Max's unified memory efficiently ran Qwen3-Coder-30B and three Smolagents without memory issues (Figure 38), with query/DataFrame/API operations in microseconds–milliseconds. Latency is slightly higher than GPU setups but suitable for edge CPS; 2025 benchmarks show MoE models like Qwen3 achieving sub-second inference in agent systems for industrial tasks[64].

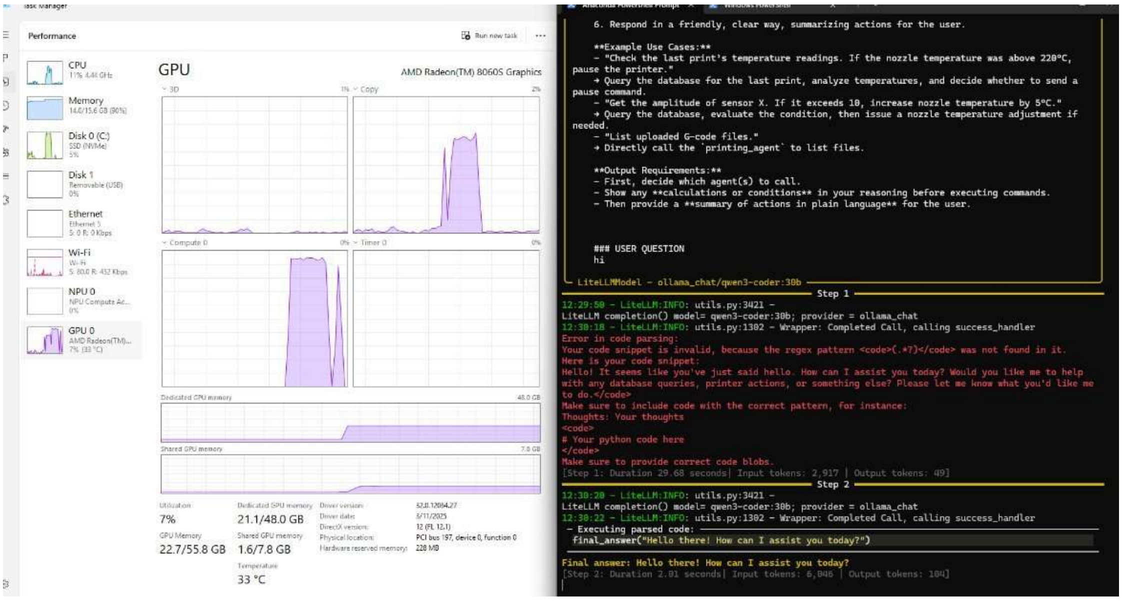


Figure 38. Performance on AMD Ryzen

6.2.4 Security

The Python sandbox limits libraries and access, while SQL queries are schema-constrained to prevent injection. Local LLM/SQLite deployment eliminates API vulnerabilities and remote connections, offering low attack surface. In 2025, LLM multi-agent systems in CPS face risks such deception attacks[63]; this setup mitigates them through isolation, aligning with guidelines for secure agentic AI in manufacturing.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This thesis has presented a novel cyber infrastructure for AI-driven analytics and control of a polymer 3D printing machine, addressing key challenges in additive manufacturing (AM) reliability and automation. By integrating multi-modal sensing (acoustic emission (AE), computer vision (CV), and thermal monitoring) with spatial defect localization, the system achieves real-time detection and mitigation of common defects such as under-extrusion, stringing, and spaghettification. The deployment of a hierarchical LLM-powered multi-agent system enables closed-loop control, natural-language querying of historical data, and conditional interventions, transforming traditional open-loop FDM printers into intelligent cyber-physical systems (CPS). A user-friendly Streamlit dashboard and modular codebase facilitate rapid deployment on existing OctoPrint setups, significantly lowering barriers for the global 3D printing community and promoting widespread adoption of advanced monitoring. Experimental results validate the system's efficacy, with CV achieving 90.2% mAP@0.50 and AE 85% accuracy, both with sub-100 ms latencies, enabling proactive quality improvements and reducing failure rates by up to 50% in unsupervised printing. This work advances Industry 4.0 by bridging sensing, data management, and AI autonomy, paving the way for scalable, efficient AM processes.

7.2 Future Work

The proposed system lays a strong foundation for AI-driven 3D printing, but several avenues for extension exist to enhance scalability, interoperability, and autonomy. Future research will focus on integrating standardized protocols, advanced simulations, and modeling frameworks to support digital twins and broader CPS applications in manufacturing.

7.2.1 Integration with MTConnect

A key future direction involves incorporating MTConnect, an open-source standard for machine tool data interchange, to enable seamless connectivity in additive manufacturing environments. MTConnect standardizes data from manufacturing equipment into XML format, facilitating real-time monitoring and predictive analytics across heterogeneous devices. By extending the current system with MTConnect adapters as shown in Figure 39, sensor data from the Ender 3 could be translated into MTConnect streams, allowing integration with enterprise systems for factory-wide oversight.

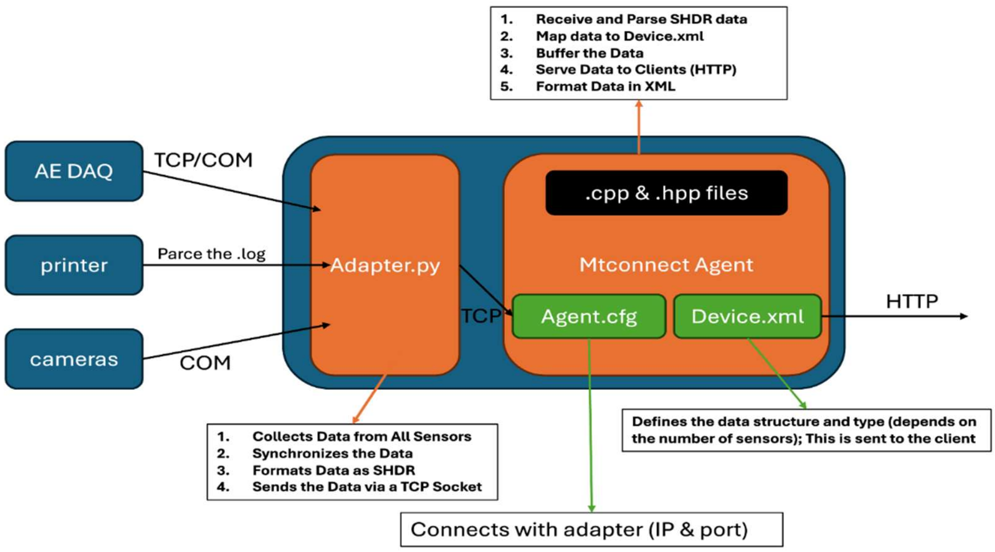


Figure 39. 3d Printing Setup Integration with MTConnect

The Additive Working Group of MTConnect is actively enhancing schemas for AM-specific items like build status and material usage, which could enable multi-printer farms with unified dashboards (see Figure 40). This would support predictive maintenance and process optimization, potentially reducing downtime by 20–30% in industrial settings, as demonstrated in MTConnect-enabled CNC and AM use cases[79].

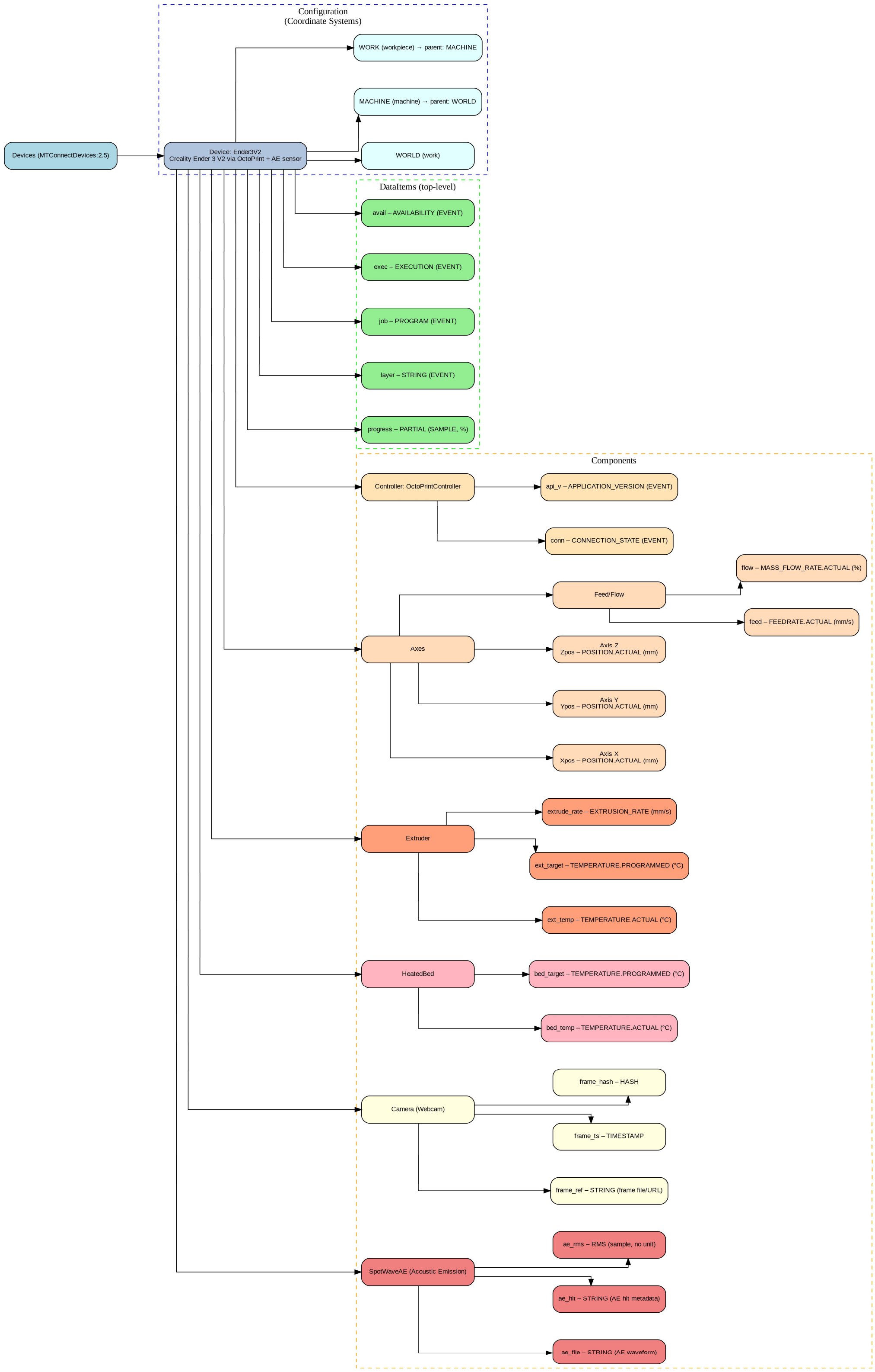

Configuration
(Coordinate Systems)
WORK (workpiece) → parent: MACHINE
MACHINE (machine) → parent: WORLD
WORLD (work)
Devices (MTConnectDevices:2.5)
Device: Ender3V2
Creality Ender 3 V2 via OctoPrint + AE sensor
DataItems (top-level)
avail – AVAILABILITY (EVENT)
exec – EXECUTION (EVENT)
job – PROGRAM (EVENT)
layer – STRING (EVENT)
progress – PARTIAL (SAMPLE, %)
Components
Controller: OctoPrintController
api_v – APPLICATION_VERSION (EVENT)
conn – CONNECTION_STATE (EVENT)
flow – MASS_FLOW_RATE.ACTUAL (%)
Feed/Flow
feed – FEEDRATE.ACTUAL (mm/s)
Axes
Axis Z
Zpos – POSITION.ACTUAL (mm)
Axis Y
Ypos – POSITION.ACTUAL (mm)
Axis X
Xpos – POSITION.ACTUAL (mm)
extrude_rate – EXTRUSION_RATE (mm/s)
Extruder
ext_target – TEMPERATURE.PROGRAMMED (°C)
ext_temp – TEMPERATURE.ACTUAL (°C)
HeatedBed
bed_target – TEMPERATURE.PROGRAMMED (°C)
bed_temp – TEMPERATURE.ACTUAL (°C)
frame_hash – HASH
Camera (Webcam)
frame_ts – TIMESTAMP
frame_ref – STRING (frame file/URL)
ae_rms – RMS (sample, no unit)
SpotWaveAE (Acoustic Emission)
ae_hit – STRING (AE hit metadata)
ae_file – STRING (AE waveform)

Figure 40. The MTConnect XML Components for the 3d Printer

7.2.2 Development of Digital Twins Using Godot and Isaac Sim

To realize the objective of digital twins for continuous learning, future work will leverage simulation engines such Godot and NVIDIA Isaac Sim for virtual replicas of the 3D printing process as illustrated in Figure 41. Godot, an open-source game engine, can model kinematic (XYZ + feedrate) and viscosity (temperature + material specs) aspects of FDM, allowing offline fine-tuning of parameters for defects like over-extrusion. Isaac Sim, powered by Omniverse, offers physics-accurate simulations for robotics and AM, integrating cuRobo for motion planning and supporting AI-driven twins for real-time optimization. By syncing real printer data (e.g., via OctoRest) with virtual models, defects can be simulated and corrections tested virtually, reducing physical trials by 50% and enabling predictive maintenance. 2025 trends emphasize AI-enhanced twins for zero-touch manufacturing, with Isaac Sim used in factory digital twins to minimize downtime[80, 81].

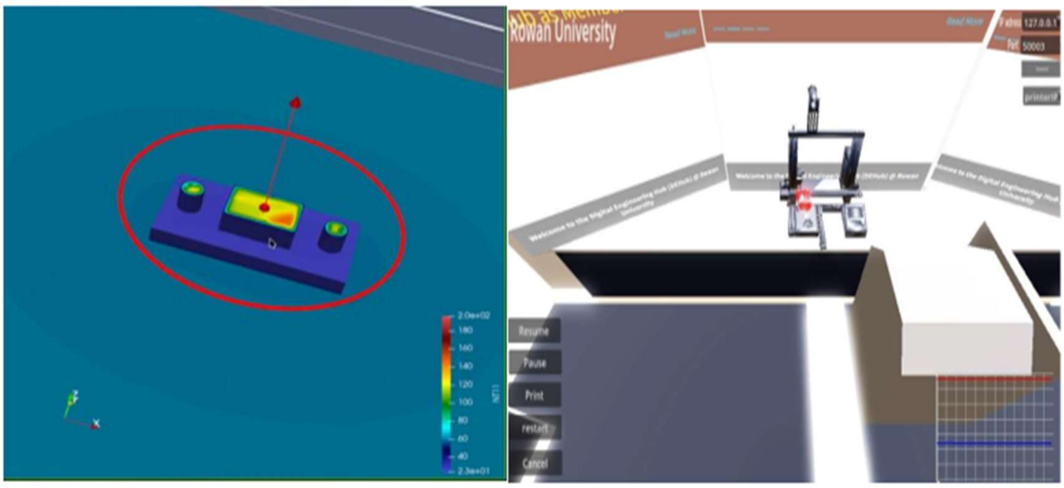


Figure 41. Digital Twin Development

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